

## Appendix I

### Mark-Houwink Parameters for Homopolymers

| Polymer     | Solvent | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|-------------|---------|--------------|---------------------------|-------|---------------------------------------|-------------|-----|
| Polystyrene | THF     | 20           | 1.55                      | 0.692 |                                       |             | 1   |
|             |         | 23           | 0.682                     | 0.766 | 50–1000                               | A           | 2   |
|             |         | 25           | 1.11                      | 0.725 |                                       | B           | 3   |
|             |         |              | 1.60                      | 0.706 | > 3                                   |             | 4   |
|             |         |              | 1.17                      | 0.725 |                                       | B           | 5   |
|             |         |              | 1.17                      | 0.717 |                                       | B           | 6   |
|             |         |              | 1.25                      | 0.707 |                                       |             | 7   |
|             |         |              | 1.68                      | 0.69  |                                       |             | 8   |
|             |         |              | 1.41                      | 0.70  |                                       |             | 9   |
|             |         |              | 1.25                      | 0.717 |                                       |             | 10  |
|             |         |              | 1.42                      | 0.70  |                                       |             | 11  |
|             |         |              | 1.16                      | 0.724 |                                       |             | 12  |
|             |         |              | 1.28                      | 0.70  |                                       |             | 13  |
|             |         |              | 0.609                     | 0.768 |                                       |             | 14  |
|             |         | 30           | 1.76                      | 0.679 |                                       |             | 15  |
|             |         |              | 0.86                      | 0.74  |                                       | B           | 16  |
|             |         |              | 1.05                      | 0.731 | 36–10 <sup>4</sup>                    |             | 17  |
|             |         |              | 1.57                      | 0.698 |                                       |             | 18  |
|             |         |              | 1.05                      | 0.731 |                                       | B           | 19  |
|             |         |              | 1.62                      | 0.701 |                                       | B           | 20  |
|             |         |              | 1.112                     | 0.723 |                                       |             | 21  |
|             |         |              | 1.28                      | 0.712 | > 10 <sup>4</sup>                     | C           | 22  |
|             |         |              | 17.1                      | 0.428 | < 10 <sup>4</sup>                     |             |     |
|             |         | 31.15        | 1.25                      | 0.717 | 4–1800                                |             | 23  |
|             |         | 35           | 1.258                     | 0.715 |                                       |             | 24  |
|             |         |              | 1.580                     | 0.704 |                                       |             | 25  |
|             |         |              | 1.60                      | 0.703 |                                       |             | 26  |
|             |         | 38           | 1.247                     | 0.720 |                                       |             | 27  |
|             |         | 40           | 1.90                      | 0.68  |                                       |             | 28  |
|             |         |              | 1.31                      | 0.719 | 19.1–1110                             | B           | 29  |
|             |         |              | 1.51                      | 0.706 | 5.11–1030                             | A           |     |
|             | benzene | 25           | 1.13                      | 0.73  |                                       |             | 30  |
|             |         | 30           | 1.623                     | 0.695 | 5–400                                 |             | 31  |
|             | toluene | 25           | 1.05                      | 0.73  |                                       |             | 32  |
|             |         |              | 0.662                     | 0.76  |                                       |             | 33  |
|             |         |              | 11.4                      | 0.470 | < 9000                                |             | 34  |
|             |         |              | 1.06                      | 0.728 | > 9000                                |             |     |

| Polymer | Solvent                | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|---------|------------------------|--------------|---------------------------|-------|---------------------------------------|-------------|-----|
| PMS     | chloroform             | 30           | 1.786                     | 0.680 | 5–2000                                |             | 31  |
|         |                        |              | 1.20                      | 0.714 | 17–10 <sup>4</sup>                    |             | 17  |
|         |                        |              | 9.86                      | 0.499 | 0.56–17                               |             |     |
|         |                        |              | 1.57                      | 0.695 |                                       | B           | 35  |
|         |                        | 60           | 0.797                     | 0.75  |                                       |             | 36  |
|         |                        | 68           | 1.528                     | 0.691 |                                       |             | 15  |
|         |                        | 25           | 0.716                     | 0.76  | 100–2800                              |             | 37  |
|         |                        | 30           | 1.36                      | 0.708 | 24–10 <sup>4</sup>                    |             | 17  |
|         |                        |              | 11.1                      | 0.502 | 0.16–24                               |             |     |
|         |                        |              | 0.49                      | 0.796 |                                       |             | 38  |
|         | <i>p</i> -dioxane      | 30           | 2.241                     | 0.655 | 5–100                                 |             | 31  |
|         | MC                     | 25           | 1.533                     | 0.695 |                                       |             | 39  |
|         |                        | 30           | 1.157                     | 0.718 |                                       |             | 40  |
|         | MC/DCA                 | 30           | 1.4                       | 0.70  |                                       | D           | 41  |
|         | cyclohexane            | 34.5         | 8.3                       | 0.50  |                                       | B           | 35  |
|         |                        | 44.5         | 2.30                      | 0.608 |                                       | B           |     |
|         | cyclohexanone          | 40           | 3.02                      | 0.643 |                                       | E           | 42  |
|         | MEK                    | 30           | 1.79                      | 0.642 |                                       | B           | 35  |
|         | DMF                    | 20           | 2.40                      | 0.63  |                                       |             | 43  |
|         |                        | 30           | 2.606                     | 0.612 | 5–400                                 |             | 31  |
|         | DMF +<br>0.01 M LiBr   | 60           | 1.30                      | 0.662 |                                       |             | 44  |
|         | DMF                    | 135          | 2.80                      | 0.606 |                                       |             | 45  |
|         | OCP                    | 25           | 14.3                      | 0.488 |                                       |             | 46  |
|         |                        | 90           | 1.2                       | 0.70  |                                       |             | 28  |
|         | TCB                    | 127          | 1.21                      | 0.71  |                                       |             | 47  |
|         |                        | 130          | 0.895                     | 0.727 |                                       | A           | 2   |
|         |                        | 135          | 1.21                      | 0.707 |                                       |             | 48  |
|         |                        |              | 1.72                      | 0.67  |                                       |             | 49  |
|         |                        | 140          | 1.90                      | 0.655 |                                       |             | 50  |
|         |                        |              | 1.34                      | 0.70  |                                       |             | 51  |
|         |                        | 145          | 4.14                      | 0.610 |                                       |             | 52  |
|         |                        |              | 3.53                      | 0.617 |                                       |             |     |
|         |                        |              | 1.75                      | 0.67  |                                       |             | 53  |
|         | ODCB                   | 87           | 1.0                       | 0.73  |                                       | B           | 54  |
|         |                        | 135          | 1.156                     | 0.715 |                                       |             | 22  |
|         |                        |              | 1.38                      | 0.70  | 2–900                                 |             | 55  |
|         |                        |              | 4.87                      | 0.606 |                                       |             | 56  |
|         |                        |              | 1.51                      | 0.693 |                                       |             | 57  |
|         |                        | 138          | 1.38                      | 0.7   |                                       | F           | 58  |
|         | <i>m</i> -cresol       | 130          | 0.846                     | 0.715 |                                       | B           | 59  |
|         |                        | 135          | 2.02                      | 0.65  | 4–200                                 |             | 28  |
|         | <i>cis</i> -decalin    | 25           | 4.0                       | 0.574 | 20–24000                              | B           | 60  |
|         | HFIP/<br>toluene (2:8) | 50           | 1.795                     | 0.657 |                                       |             | 61  |
|         | Phenol/TCB (1:1)       | 135          | 1.303                     | 0.69  |                                       |             | 62  |
|         | CNP                    | 208          | 1.86                      | 0.657 |                                       |             | 63  |
|         | THF                    | 25           | 4.2                       | 0.608 |                                       |             | 11  |
|         | TCB                    | 135          | 1.61                      | 0.672 |                                       |             | 48  |

| Polymer | Solvent                                                                           | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$    | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|---------|-----------------------------------------------------------------------------------|--------------|---------------------------|--------|---------------------------------------|-------------|-----|
| PVC     | THF                                                                               | 25           | 1.63                      | 0.766  | 20-170                                |             | 4   |
|         |                                                                                   |              | 1.50                      | 0.77   | 20-100                                | B           | 6   |
|         |                                                                                   |              | 7.2                       | 0.61   | 27-100                                |             | 8   |
|         |                                                                                   |              | 4.48                      | 0.70   |                                       |             | 14  |
|         |                                                                                   |              | 1.6                       | 0.77   |                                       | G           | 64  |
|         |                                                                                   |              | 4.76                      | 0.673  |                                       |             | 65  |
|         |                                                                                   | 30           | 1.60                      | 0.77   | 100-1000                              | B           | 5   |
|         |                                                                                   | 35           | 1.556                     | 0.690  |                                       |             | 24  |
|         |                                                                                   |              | 1.8                       | 0.76   |                                       | B           | 66  |
|         |                                                                                   | 110          | 1.61                      | 0.762  |                                       |             | 67  |
| PVAc    | TCB                                                                               |              |                           |        |                                       |             |     |
|         | THF                                                                               | 25           | 3.50                      | 0.63   | 10-1000                               | B           | 5   |
|         |                                                                                   |              | 0.942                     | 0.737  |                                       |             | 12  |
|         |                                                                                   |              | 0.51                      | 0.791  |                                       |             | 68  |
|         |                                                                                   |              | 1.6                       | 0.70   |                                       |             | 69  |
|         |                                                                                   | 30           | 1.49                      | 0.698  |                                       |             | 70  |
|         |                                                                                   | 35           | 1.56                      | 0.708  |                                       |             | 71  |
| PMMA    | TCB                                                                               | 135          | 1.37                      | 0.687  |                                       |             | 70  |
|         | THF                                                                               | 23           | 0.93                      | 0.69   |                                       |             | 8   |
|         |                                                                                   | 25           | 1.28                      | 0.69   | 150-1200                              |             | 9   |
|         |                                                                                   |              | 0.958                     | 0.695  |                                       |             | 13  |
|         |                                                                                   |              | 0.787                     | 0.75   |                                       |             | 65  |
|         |                                                                                   |              | 1.48                      | 0.677  |                                       |             | 72  |
|         |                                                                                   |              | 1.99                      | 0.660  |                                       |             | 73  |
|         |                                                                                   | 35           | 1.22                      | 0.69   |                                       |             | 26  |
|         | chloroform                                                                        | 25           | 1.47                      | 0.714  |                                       |             | 73  |
|         |                                                                                   | 30           | 0.43                      | 0.80   |                                       |             | 38  |
|         | acetone                                                                           | 25           | 3.66                      | 0.578  |                                       |             | 73  |
|         | MEK                                                                               | 25           | 0.71                      | 0.72   |                                       |             | 38  |
|         | benzene                                                                           | 25           | 0.55                      | 0.76   |                                       |             | 30  |
|         |                                                                                   |              | 4.38                      | 0.597  |                                       |             | 73  |
|         |                                                                                   | 30           | 0.52                      | 0.76   |                                       |             | 74  |
|         | toluene                                                                           | 25           | 1.91                      | 0.68   |                                       |             | 33  |
|         |                                                                                   |              | 2.08                      | 0.64   | > 7800                                |             | 34  |
|         | cyclohexanone                                                                     | 40           | 3.66                      | 0.603  |                                       | E           | 42  |
|         | DMF                                                                               | 25           | 0.404                     | 0.787  |                                       | A           | 75  |
|         |                                                                                   |              | 2.5                       | 0.625  |                                       | B           |     |
|         | TFE                                                                               | 30           | 0.281                     | 0.88   |                                       |             | 76  |
|         | (100% isotactic)<br>(92% iso)<br>(83% iso)<br>(35% iso)<br>(23% iso)<br>(15% iso) | 25           | 1.665                     | 0.660  |                                       |             | 77  |
|         |                                                                                   |              | 1.690                     | 0.659  |                                       |             |     |
|         |                                                                                   |              | 1.719                     | 0.658  |                                       |             |     |
|         |                                                                                   |              | 0.984                     | 0.692  |                                       |             |     |
|         |                                                                                   |              | 0.9577                    | 0.695  |                                       |             |     |
| PEMA    | THF                                                                               | 25           | 0.8892                    | 0.694  |                                       |             |     |
|         |                                                                                   |              |                           |        |                                       |             | 14  |
| PBMA    | THF                                                                               | 25           | 1.549                     | 0.679  |                                       |             | 78  |
|         |                                                                                   |              | 2.25                      | 0.66   |                                       |             |     |
|         | MEK                                                                               | 25           | 0.503                     | 0.758  |                                       |             | 14  |
|         |                                                                                   |              | 0.613                     | 0.7258 | $10^4-2 \times 10^4$                  |             | 79  |
|         |                                                                                   |              | 0.970                     | 0.68   | $10^5-7 \times 10^5$                  |             | 80  |

| Polymer         | Solvent | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref       |
|-----------------|---------|--------------|---------------------------|-------|---------------------------------------|-------------|-----------|
| POMA            | THF     | 25           | 5.556                     | 0.560 |                                       |             | 14        |
| PLMA            | THF     | 25           | 0.73                      | 0.69  |                                       |             | 81        |
| PSMA            | THF     | 30           | 0.90                      | 0.67  |                                       | B           | 82        |
| PMA             | THF     | 25           | 0.388                     | 0.82  |                                       |             | 83        |
|                 | toluene | 35           | 2.1                       | 0.60  |                                       |             | 84        |
| PtBA            | THF     | 25           | 0.33                      | 0.80  |                                       | H           | 85        |
| Polyethylene    | ODCB    | 135          | 4.77                      | 0.70  | 6–700                                 |             | 56        |
|                 |         |              | 4.90                      | 0.74  |                                       |             | 86        |
|                 |         | 138          | 5.06                      | 0.7   |                                       | F           | 58        |
|                 | TCB     | 135          | 5.50                      | 0.704 |                                       |             | 87        |
|                 |         |              | 4.34                      | 0.724 |                                       |             | 48        |
|                 |         |              | 9.54                      | 0.64  |                                       |             | 49        |
|                 |         |              | 5.3                       | 0.7   |                                       | F           | 88        |
| HDPE            | ODCB    | 135          | 5.05                      | 0.693 | 10–1000                               |             | 57        |
|                 | TCB     | 127          | 3.58                      | 0.74  |                                       |             | 47        |
|                 |         | 130          | 3.92                      | 0.725 |                                       |             | 89        |
|                 |         | 135          | 9.54                      | 0.64  |                                       |             | 90        |
|                 |         |              | 5.26                      | 0.70  |                                       |             | 88        |
|                 |         |              | 5.23                      | 0.70  |                                       |             | 91        |
|                 |         |              | 7.1                       | 0.67  |                                       |             | 92        |
|                 |         |              | 5.63                      | 0.70  |                                       |             | 93        |
|                 |         | 140          | 3.95                      | 0.726 |                                       |             | 50        |
|                 |         |              | 3.23                      | 0.735 |                                       |             | 94        |
|                 |         | 145          | 1.73                      | 0.784 |                                       |             | 52        |
|                 | decalin | 135          | 4.17                      | 0.71  |                                       |             | 95        |
|                 |         |              | 3.9                       | 0.74  |                                       |             | 96        |
|                 |         |              | 4.6                       | 0.73  |                                       |             | 91        |
|                 |         | 138          | 6.2                       | 0.7   |                                       | F           | 97        |
|                 | decane  | 120          | 6.98                      | 0.65  |                                       |             | 95        |
| LDPE            | ODCB    | 135          | 5.06                      | 0.70  | 0.2–200                               |             | 99        |
| LLDPE           | TCB     | 145          | 5.91                      | 0.69  |                                       |             | 53        |
| Hydrogenated PB | TCB     | 135          | 2.7                       | 0.746 | 2.4–40                                |             | 58        |
| PP              | TCB     | 135          | 1.37                      | 0.75  |                                       |             | 48        |
|                 |         |              | 1.76                      | 0.73  |                                       |             | 91        |
|                 |         | 145          | 1.56                      | 0.76  |                                       |             | 53        |
|                 | ODCB    | 135          | 1.0                       | 0.78  |                                       |             | 86        |
|                 |         |              | 2.42                      | 0.707 |                                       | B           | 98        |
|                 |         |              | 1.10                      | 0.80  |                                       |             | 99        |
|                 | decalin | 135          | 1.10                      | 0.80  |                                       |             | 86,<br>91 |
| Polyisobutylene | THF     | 25           | 5                         | 0.60  |                                       | H           | 100       |
|                 |         | 40           | 6.47                      | 0.593 | 0.73–67.6                             | A           | 29        |
|                 |         |              | 5.79                      | 0.593 | 0.82–80.4                             | B           |           |
| Polycarbonate   | THF     | 25           | 3.99                      | 0.77  |                                       | I           | 28        |
|                 |         |              | 4.9                       | 0.67  | 7–270                                 |             | 101       |
|                 |         |              | 4.12                      | 0.69  | 10–300                                | J           | 102       |
|                 |         | 35           | 3.18                      | 0.680 |                                       | K           | 25        |
| PB              | toluene | 30           | 5.19                      | 0.679 | 0.5–10 <sup>4</sup>                   |             | 17        |
|                 |         |              | 2.888                     | 0.734 |                                       |             | 21        |
|                 |         | 68           | 2.103                     | 0.753 |                                       |             | 15        |

| Polymer                      | Solvent                | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$    | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|------------------------------|------------------------|--------------|---------------------------|--------|---------------------------------------|-------------|-----|
|                              | THF                    | 25           | 1.012                     | 0.621  |                                       |             | 27  |
|                              |                        |              | 18.4                      | 0.58   |                                       |             | 65  |
|                              |                        |              | 1.6                       | 0.776  |                                       |             | 67  |
|                              |                        | 30           | 4.57                      | 0.693  | 0.5–10 <sup>4</sup>                   |             | 17  |
|                              |                        |              | 2.56                      | 0.74   |                                       | B           | 82  |
|                              |                        | 40           | 3.96                      | 0.697  |                                       | L           | 103 |
|                              | chloroform             | 30           | 4.51                      | 0.704  | 0.19–10 <sup>4</sup>                  |             | 17  |
|                              |                        | 40           | 5.78                      | 0.67   | 10–100                                |             | 28  |
|                              |                        | 30           | 76.0                      | 0.44   |                                       |             |     |
| ( <i>cis/trans</i> = 0.8)    |                        |              | 25                        | 2.36   | 0.75                                  | 3–6         |     |
| (20% <i>cis</i> -20% vinyl)  |                        |              | 4.57                      | 0.693  | 80–1100                               |             | 10  |
| (8% vinyl)                   |                        |              | 4.51                      | 0.693  | 20–300                                |             |     |
| (28% vinyl)                  |                        |              | 4.28                      | 0.693  | 20–200                                |             |     |
| (52% vinyl)                  |                        |              | 4.03                      | 0.693  | 20–200                                |             |     |
| (73% vinyl)                  |                        |              | 4.10                      | 0.693  | 2.4–40                                |             |     |
| SBR(25% styrene)             | THF                    | 25           | 4.10                      | 0.693  | 70–1000                               |             | 28  |
|                              |                        | 40           | 3.18                      | 0.70   |                                       |             |     |
| Natural rubber               |                        |              |                           |        |                                       |             |     |
| (poly- <i>cis</i> -isoprene) | THF                    | 25           | 1.09                      | 0.79   | 10–1000                               | G           |     |
| Polyisoprene                 | THF                    | 25           | 1.77                      | 0.753  | 40–500                                |             | 10  |
| Butyl rubber                 | THF                    | 25           | 0.85                      | 0.75   | 4–4000                                |             | 28  |
| PDSX                         | toluene                | 25           | 0.828                     | 0.72   |                                       |             | 38  |
|                              |                        | 60           | 0.977                     | 0.725  |                                       |             | 36  |
|                              | chloroform             | 30           | 0.54                      | 0.77   |                                       |             | 38  |
|                              | cyclohexane            | 35           | 0.63                      | 0.77   |                                       |             | 104 |
|                              | ODCB                   | 87           | 8.19                      | 0.50   | 20–800                                |             | 54  |
|                              |                        | 138          | 3.83                      | 0.57   | 25–300                                |             | 55  |
|                              | DMAC                   | 25           | 1.3                       | 0.69   |                                       | M           | 105 |
| Nylon 6                      | <i>m</i> -cresol       | 25           | 32                        | 0.62   | 0.5–5                                 |             | 28  |
|                              | OCP                    | 90           | 6.2                       | 0.64   |                                       |             |     |
|                              | HFB                    | 25           | 4.2                       | 0.76   |                                       |             | 106 |
|                              | TFE                    | 25           | 4.58                      | 0.742  | 20–100                                |             | 107 |
|                              |                        |              | 5.36                      | 0.75   | 13–100                                |             | 108 |
| Nylon 66                     | <i>m</i> -cresol       | 25           | 3.53                      | 0.79   | 0.15–50                               |             | 28  |
|                              |                        | 130          | 4.01                      | 1.00   | 8–24                                  | B           | 59  |
|                              | HFIP                   |              | 19.8                      | 0.63   |                                       |             | 109 |
| Nylon 610                    | <i>m</i> -cresol       | 25           | 1.35                      | 0.96   | 8–24                                  |             | 28  |
| Nylon 11                     | <i>m</i> -cresol       | 30           | 9.1                       | 0.69   |                                       |             | 110 |
| Nylon 12                     | <i>m</i> -cresol       | 25           | 4.6                       | 0.75   | 10–125                                |             | 28  |
|                              | HFIP/<br>toluene (2:8) | 50           | 6.163                     | 0.6585 |                                       |             | 61  |
| Trogamid T                   | DMF                    | 22           | 2.737                     | 0.706  |                                       |             | 111 |
| TFA-nylon 6                  | MC                     | 25           | 3.355                     | 0.64   |                                       | N           | 112 |
|                              | THF                    |              | 1.66                      | 0.70   |                                       | O           | 113 |
| TFA-nylon 12                 | MC                     | 30           | 0.738                     | 0.81   |                                       | P           | 114 |
| TFA-Trogamid                 | THF                    | 25           | 0.87                      | 0.73   |                                       | O           | 113 |
| PET                          | <i>m</i> -cresol       | 25           | 0.077                     | 0.95   |                                       |             | 28  |
|                              |                        | 135          | 1.75                      | 0.81   | 27–32                                 |             |     |
|                              |                        |              | 2.0                       | 0.90   | 0.45–0.8                              |             |     |
|                              | OCP                    | 23           | 6.31                      | 0.658  |                                       |             | 115 |
|                              | PTCE                   | 23           | 7.44                      | 0.648  |                                       |             |     |

| Polymer          | Solvent          | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|------------------|------------------|--------------|---------------------------|-------|---------------------------------------|-------------|-----|
|                  | HFIP             | 23           | 5.20                      | 0.695 |                                       |             |     |
|                  | PFP              | 23           | 3.85                      | 0.723 |                                       |             |     |
|                  | HFIP/PFP (1:1)   | 23           | 4.50                      | 0.705 |                                       |             |     |
|                  |                  | 25           | 6.31                      | 0.658 |                                       |             | 46  |
|                  | MC/DCAA          | 30           | 4.5                       | 0.68  |                                       | Q           | 41  |
|                  | MC/HFIP (7:3)    | 25           | 4.034                     | 0.691 |                                       | R           | 116 |
| LCA polyester    | PFP              | 50           | 0.9205                    | 1.05  |                                       |             | 117 |
| PTHF             | cyclohexanone    | 40           | 4.16                      | 0.676 |                                       | E           | 42  |
| Polyoxymethylene | DMF              | 135          | 0.932                     | 0.79  |                                       |             | 45  |
| PVDBC            | THF              | 25           | 9.1                       | 0.827 |                                       |             | 118 |
| PVB              | THF              | 25           | 1.4                       | 0.80  |                                       | H           | 119 |
| POA              | THF              | 25           | 0.842                     | 0.202 | > 12.5                                | G           | 120 |
|                  |                  |              | 0.961                     | 0.674 | < 12.5                                | G           |     |
| PDEPP            | THF              | 25           | 2.5                       | 0.65  |                                       | S           | 121 |
| PDPPP, PAPP      | THF              | 30           | 1.19                      | 0.649 |                                       | B, T        | 20  |
| PVK              | THF              | 25           | 0.127                     | 0.80  |                                       | U           | 122 |
| PPS              | CNP              | 210          | 0.891                     | 0.747 |                                       |             | 63  |
| PEEK             | sulfuric acid    | 25           | 0.3849                    | 0.94  |                                       | B           | 62  |
|                  | phenol/TCB (1:1) | 115          | 7.588                     | 0.67  |                                       | B           | 62  |

**Abbreviations.** THF – tetrahydrofuran; MC – methylene chloride; MEK – methyl ethyl ketone; DMF – *N,N*-dimethylformamide; OCP – *o*-chlorophenol; TCB – 1, 2,4-trichlorobenzene; ODCB – *o*-dichlorobenzene; HFIP – hexafluoroisopropanol; CNP – 1-chloronaphthalene; TFE – trifluoroethanol; DMAC – *N,N*-dimethylacetamide; HFB – hexafluorobutanol; PTCE – phenol/tetrachloroethane (6:4); PFP – pentafluorophenol; PMS – poly(methyl styrene); PVC – poly(vinyl chloride); PVAc – poly(vinyl acetate); PMMA – poly(methyl methacrylate); PEMA – poly(ethyl methacrylate); PBMA – poly(butyl methacrylate); POMA – poly(octyl methacrylate); PLMA – poly(*n*-lauryl methacrylate); PSMA poly(stearyl methacrylate); PMA – poly(methyl acrylate); PtBA – poly(*tert*-butyl acrylate); HDPE – high density polyethylene (PE); LDPE – low density PE; LLDPE – linear LDPE; PP – polypropylene; PB – polybutadiene; SBR – styrene-butadiene rubber; PDSX – polydimethylsiloxane; Trogamid T – poly(trimethylhexamethylene terephthalamide); PET – poly(ethylene terephthalate); LCA polyester-liquid crystalline aromatic polyester; PTHF – poly(tetrahydrofuran); PVDBC – poly(*N*-vinyl-3,6-dibromocarbazole); PVB – poly(vinyl butyral); POA – poly[*N*-(*n*-octadecyl) maleimide]; PDEPP – poly(diethoxyphosphazene); PDPPP – poly(diphenyl phosphazene); PAPP – poly(aryloxy phosphazene); PVK – poly(*N*-vinyl-3,6-dibromocarbazole); PPS – poly(phenylene sulfide); PEEK – poly(arylether ether ketone).

**Remarks.** A –  $[\eta] = KM_n^a$ ; B –  $[\eta] = KM_w^a$ ; C – narrow MW distribution; D – MC/dichloroacetic acid (8:2) + 0.01 M tetrabutylammonium acetate; E – 0.01 M tetrabutylammonium nitrate included; F –  $[\eta] = KM_n^a$ ; G – use  $K$  and  $a$  from [4] for PS; H – use  $K$  and  $a$  from [6] for PS; I – 4,4'-dihydroxy-2,2-propane carbonate; J – use  $K$  and  $a$  from [9] for PS; K – bisphenol A/diethylene glycol (50:50); L – 35% cis, 55% trans, 10% 1,2; M – poly(methyl) (pyridine-3-yl) siloxane; N – use  $K$  and  $a$  from [37] for PS; O – use  $K = 1.31 \times 10^{-2}$  and  $a = 0.714$  for PS; P – use  $K$  and  $a$  from [38] for PS; Q – MC/dichloroacetic acid + 0.01 M tetrabutylammonium acetate; R – use  $K = 7.998 \times 10^{-2}$  and  $a = 0.54$  for PS; S – 0.1% tetra(*n*-butylammonium) bromide included; T – 0.1 M LiBr included; U – use  $K$  and  $a$  from [3] for PS.

## References

1. ECHARRI J, IRUIN JJ, GUZMAN GM, AMSORENA J (1979) *Makromol Chem* 180:2749
2. BONI KA, SLIEMERS FA, STICKNEY PB (1968) *J Polym Sci Part A* 26:1579
3. SPATORICO AL, COULTER B (1973) *J Polym Sci Polym Phys Ed* 11:1139
4. PROVDER T, ROSEN EM (1970) *Sep Sci* 5:437
5. GOEDHAR D, OPSHOOR A (1970) *J Polym Sci Part A-2*, 8:1227
6. KOLÍNSKY M, JANČA J (1974) *J Polym Sci Polym Chem Ed* 12:1181
7. MORRIS MC (1971) *J Chromatogr* 55:203
8. GRUBISIC Z, REMPP P, BENOIT H (1967) *J Polym Sci Part B* 5:753
9. RUDIN A, HOEGY HW (1972) *J Polym Sci Part A-1*, 10:217
10. KRAUS C, STACY CJ (1972) *J Polym Sci Part A-2*, 10:657
11. SCHRODER UKO, EBERT KH (1987) *Makromol Chem* 188:1415
12. COLEMAN TA, DAWKINS JV (1986) *J Liq Chromatogr* 9:1191
13. JENKINS R, PORTER RS (1980) *J Polym Sci Polym Lett Ed* 18:743
14. SAMAY G, KUBÍN M, PODEŠVA J (1978) *Angew Makromol Chem* 72:185
15. AMBLER MR (1973) *J Polym Sci Polym Chem Ed* 11:191
16. WILLIAMS RC, SCHMIT JA (1971) *J Polym Sci Part B* 9:413
17. AMBLER MR (1980) *J Appl Polym Sci* 25:901
18. SUDDABY KG, SANAYEI RA, O'DRISCOLL KF, RUDIN A (1993) *Makromol Chem* 194:1965
19. AMBLER MR, MCINTYRE D (1977) *J Appl Polym Sci* 21:2269
20. DEJAEGER R, LECACHEUX D, POTIN PH (1990) *J Appl Polym Sci* 39:1793
21. SMITH WV (1974) *J Appl Polym Sci* 18:3685
22. HANEY MA, ARMONAS JE, ROSEN L (1987) In: PROVDER T (ed) *ACS Symposium Series No. 352*, ACS, Washington, p 119
23. VAN DIJK JAPP, SMIT JAM, KOHN FE, FEIJEN J (1983) *J Polym Sci Polym Chem Ed* 21:197
24. PERKINS G, HAEHN (1990) *J Vinyl Technol* 12:2
25. MAHABADI HKH (1989) *International GPC Symposium*, Newton, Mass, 434
26. MAHABADI HKH (1985) *J Appl Polym Sci* 30:1535
27. ITO K, SAITO T, AOYAMA T (1987) *Polymer* 28:1589
28. EVANS JM (1973) *Polym Eng Sci* 13:401
29. XIE J (1994) *Polymer* 35:2385
30. TSITSILIANIS C, MITSIANI G, DONDOS A (1989) *J Polym Sci Part B Polym Phys* 27:763
31. ZINBO M, PARSONO JL (1971) *J Chromatogr* 55:55
32. WINTERMANTEL M, ANTONIETTI M, SCHMIDT M (1993) *J Appl Polym Sci, Appl Polym Symp* 52:91
33. PRICE GJ, MOORE JW, GUILLET JE (1989) *J Polym Sci Part A Polym Chem* 27:2925
34. SANAYEI RA, O'DRISCOLL KF (1991) *J Macromol Sci Chem A* 28:987
35. VARMA BK, FUJITA Y, TAKAHASHI M, NOSE T (1984) *J Polym Sci Polym Phys Ed* 22:1781
36. MANDIK L, FOKSOVÁ A, FOLTYN J (1979) *J Appl Polym Sci* 24:395
37. OTH J, DESREUX V (1954) *Bull Soc Chim Belges* 63:285
38. DAWKINS JV (1968) *J Macromol Sci Phys B* 2:623
39. BIAGINI E, COSTA G, GATTIGLIA E, IMPERATO A, RUSSO S (1987) *Polymer* 28:114
40. OGAWA T, SAKAI M (1988) *J Polym Sci Part A Polym Chem* 26:3141
41. MOUREY TH, BRYAN TG, GREENER J (1993) *J Chromatogr A* 657:377
42. MOUREY TH, MAILLER SM, FERRAR WT, MOLAIRE TR (1989) *Macromolecules* 22:4286
43. KRANZ D, POHL HU, BAUMANN H (1972) *Angew Makromol Chem* 26:67
44. AZUMA C, DIAS ML, MANO EB (1986) *Macromol Chem Macromol Symp* 2:169
45. OGAWA T (1990) *J Liq Chromatogr* 13:51
46. MARTIN L, LAVINE M, BALKE ST (1992) *J Liq Chromatogr* 15:1817
47. SCHEINERT W (1977) *Angew Makromol Chem* 63:117
48. COLL H, GILDING DK (1970) *J Polym Sci Part A-2* 6:1579
49. WILLIAMSON GR, CERVENKA A (1972) *Eur Poly J* 8:1009
50. BARLOW A, WILD L, RANGANATH R (1977) *J Appl Polym Sci* 21:3319

51. KAŠPARKOVA V, OMMUNDSEN E (1993) *Polymer* 34:1765
52. LEW R, CHEUNG P, BALKE ST, MOUREY TH (1993) *J Appl Polym Sci* 47:1685
53. GRINSHUPUN V, RUDIN A (1985) *Makromol Chem Rapid Commun* 6:219
54. DAWKINS JV, HEMMING M (1972) *Makromol Chem* 155:75
55. DAWKINS JV, MADDOCK JW, COUPE DJ (1970) *J Polym Sci Part A-2* 8:1803
56. TROTH HG (1968) 5th International Seminar for GPC
57. POLLOCK DJ, KRATZ RF (1968) 6th International Seminar for GPC
58. DAWKINS JV, MADDOCK JW (1971) *Eur Polym J* 7:1537
59. DUDLEY MA (1972) *J Appl Polym Sci* 16:493
60. KULICKE WM, PRESCHER M (1984) *Makromol Chem* 185:2619
61. OGAWA T, SAKAI M, ISHITOBI W (1985) *J Polym Sci Polym Chem Ed* 24:109
62. DEVAUX J, DELIMOY D, DAOUST D, LEGRAS R, MERCIER JP (1985) *Polymer* 26:1994
63. STACY CJ (1986) *J Appl Polym Sci* 34:3959
64. LEDERER K, AMTMANN I, VIJAYAKUMAR S, BILLIANI J (1990) *J Liq Chromatogr* 13:1849
65. CHAPLIN RP, HAKEN JK, PADDON JJ (1979) *J Chromatogr* 171:55
66. HATTORI S, ENDOH H, NAKAHARA H, KAMATA T, HAMASHIMA M (1978) *Polym J* 10:173
67. PANG S, RUDIN A (1993) *J Appl Polym Sci* 49:1189
68. PARK WS, GRAESSLEY WW (1977) *J Polym Sci Polym Phys Ed* 15:71
69. CANE F, CAPACCIOLI T (1978) *Eur Polym J* 14:185
70. LECACHEUX D, LESEC J, QUIVORON C, PRECHNER R, PANARAS R, BENOIT H (1984) *J Appl Polym Sci* 29:1569
71. ATKINSON CML, DIETR R (1979) *Eur Polym J* 15:21
72. CHEE KK (1985) *J Appl Polym Sci* 30:1323
73. DOBBIN CJB, RUDIN A, TCHIR MF (1982) *J Appl Polym Sci* 27:1081
74. COHN-GINSBERG E, FOX TG, MASON HF (1962) *Polymer* 3:97
75. KOSSLER I, NETOPILIK M, SCHULZ G, GNAUCK R (1982) *Polym Bull* 7:597 (1982)
76. VEITH CA, COHEN RE (1989) *Polymer* 30:942
77. JENKINS R, PORTER RS (1982) *Polymer* 23:105
78. GRUBISIC-GALLOT Z, LINGELSER JP, GALLOT Y (1990) *Polym Bull* 23:389
79. SIMIONESCU CI, SIMIONESCU BC, NEAMTU I, IOAN S (1987) *Polymer* 28:165
80. FEDORS RF (1979) *Polymer* 20:226
81. MAHABADI HKH, O'DRISCOLL KH (1977) *J Appl Polym Sci* 21:1283
82. ZHONGDE X, MINGSHI S, HADJICHRISTIDIS N, FETTERS LJ (1981) *Macromolecules* 14:1591
83. SZESZTAY M, TUEDOES F (1981) *Poly Bull* 5:429
84. KRAUNAKARAN K, SANTAPPA M (1968) *J Polym Sci Part A-2*, 6:713
85. MRKVIČKOVA L, DANHELKA J (1990) *J Appl Polym Sci* 41:1929
86. OGAWA T, INABA T (1977) *J Appl Polym Sci* 21:2979
87. LEHTINEN A, VAINIKKA R (1989) International GPC Symposium, Newton, MA, 612
88. PEYROUSET A, PRECHNER R, PANARIS R, BENOIT H (1975) *J Appl Polym Sci* 19:1363
89. WAGNER HL, HOEVE CAJ (1973) *J Polym Sci Polym Phys Ed* 11:1189
90. WILLIAMSON GR, CERVENKA A (1972) *Eur Polym J* 8:1009
91. CROUZET P, MARTENS A, MANGIN P (1971) *J Chromatogr Sci* 9:525
92. COTE JA, SHIDA M (1971) *J Polym Sci Part A-2*, 9:421
93. HERT M, STRAZIELLE C (1983) *Makromol Chem* 184:135
94. WILD L, RANGANATH R, RYLE T (1971) *J Polym Sci Part A-2*, 9:2137
95. BOHM LL, LANVER U, LECHNER MD (1983) *Makromol Chem* 184:585
96. DE LA CUESTA MO, BILLMEYER FW Jr (1963) *J Polym Sci A-1*, 1:1721
97. CHIANG R (1965) *J Phys Chem* 69:1645
98. ATKINSON CML, DIETZ R (1976) *Makromol Chem* 177:213
99. OGAWA T, SUZUKI Y, INABA T (1972) *J Polym Sci Part A-1*, 10:737
100. MRKVIČKOVA L, LOPOUR P, POKORNY S, JANČA J (1980) *Angew Makromol Chem* 90:217
101. MOORE WR, UDDIN M (1969) *Eur Polym J* 5:185
102. BAILLY C, DAOUST D, LEGRAS R, MERCIER JP, STRAZIELLE C, LAPP A (1986) *Polymer* 27:1410



103. BARLOW A, WILD L, ROBERTS T (1971) *J Chromatogr* 55:155
104. DAWKINS JV, HEMMING M (1975) *Makromol Chem* 176:1777
105. VILENCHIK LZ, RUBINSTAJN S, ZELDIN M, FIRE WK (1991) *J Polym Sci Part B Polym Phys* 29:1137
106. COSTA G, RUSSIO S (1982) *J Macromol Sci Chem Ed* A18:299
107. AHARONI SM, CILURSO FG, HANRAHALN JM (1985) *J Appl Polym Sci* 30:2505
108. MATTIUSSI A, GECELE GB, FRANCESCONI R (1969) *J Polym Sci A-2*, 7:411
109. SAMANTA SR (1992) *J Appl Polym Sci* 45:1635
110. TERAN CC, MACCHI EM, FIGINI RV (1987) *J Appl Polym Sci* 34:2433
111. HEROLD J, MEYERHOFF G (1980) *Makromol Chem* 181:2625
112. BIAGINI E, COSTA G, GATTIGLIA E, IMPERAT A, RUSSO S (1987) *Polymer* 28:114
113. WEISSKOPF K (1985) *Polymer* 26:1187
114. OGAWA T, SAKAI M (1988) *J Polym Sci Part A Polym Chem* 26:3141
115. BERKOWITZ S (1984) *J Appl Polym Sci* 29:4353
116. OVERTON JR, BROWING HL Jr (1984) In: PROVDER T (ed) *ACS Symposium Series No. 245*, ACS, Washington, p 219
117. KINUGAWA A (1991) *J Liq Chromatogr* 14:1315
118. HORTA A, Saíz E, BARRALES-RIENDA JM, GÓMEZ PAG (1986) *Polymer* 27:139
119. MRKVIČKOVA L, DANHELKA J, POKORY S (1984) *J Appl Polym Sci* 29:803
120. BARRALES-RIENDA JM, GALIECA CR, HORTA A (1983) *Macromolecules* 16:932
121. TARAZONA MP, BRAVO J, RODRIGO MM, SAIZ E (1991) *Polym Bull* 26:465
122. BARRALES-RIENDA JM, GOMEZ PAG, HORTA A, SAIZ E (1985) *Macromolecules* 18:2572

## Appendix II

### Mark-Houwink Parameters for Polar Polymers

| Polymer        | Solvent      | Temp<br>(°C)                          | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|----------------|--------------|---------------------------------------|---------------------------|-------|---------------------------------------|-------------|-----|
| PAN            | DMF          | 20                                    | 3.25                      | 0.725 |                                       |             | 1   |
|                |              | 25                                    | 2.43                      | 0.75  |                                       |             | 2   |
|                |              | 60                                    | 0.146                     | 0.897 |                                       | A           | 3   |
| PVP            | DMF          |                                       | 2.12                      | 0.75  |                                       | A           | 4   |
|                |              | 60                                    | 1.85                      | 0.646 |                                       | A           | 3   |
|                |              | MeOH/H <sub>2</sub> O (1:1)           | 25                        | 0.851 |                                       | B           | 5   |
| P2VP           | THF          | 25                                    | 2.23                      | 0.66  |                                       |             | 6   |
|                |              | 35                                    | 3.02                      | 0.61  |                                       |             | 7   |
|                |              | TFA + NaNO <sub>3</sub>               | 35                        | 0.575 | 0.922                                 | C           |     |
| Polyacrylamide | water        |                                       | 1.62                      | 0.745 |                                       | D           |     |
|                |              |                                       | 2.57                      | 0.651 |                                       | E           |     |
|                |              | pyridine                              | 25                        | 0.445 | 0.78                                  | F           | 8   |
|                |              | NMP                                   | 80                        | 0.88  | 0.73                                  | G           |     |
|                |              | DMF                                   | 50                        | 1.23  | 0.69                                  | H           |     |
|                |              | sodium formate                        | 25                        | 0.25  | 0.93                                  | I           | 9   |
|                |              |                                       |                           | 0.50  | 0.83                                  | J           |     |
|                |              |                                       |                           | 0.17  | 0.95                                  | K           |     |
|                |              | see Remark                            | 25                        | 2.5   | 0.68                                  | L           | 10  |
|                |              |                                       | 25                        | 6.80  | 0.66                                  |             | 11  |
|                |              |                                       |                           | 1.0   | 0.755                                 | M           | 12  |
|                |              |                                       | 30                        | 0.631 | 0.80                                  |             | 13  |
|                |              | 0.1 M Na <sub>2</sub> SO <sub>4</sub> | 25                        | 1.94  | 0.70                                  |             | 14  |
|                |              | 0.5 M NaCl                            | 25                        | 1.21  | 0.746                                 | pH 9, M     | 15  |
|                |              | 1 M NaCl                              |                           | 1.84  | 0.72                                  | pH 9, M     |     |
| NaPA           | 0.3 N NaCl   | 25                                    | 1.69                      | 0.75  |                                       |             | 16  |
| Phenol (RN)    | acetone      | 25                                    | 4.7                       | 1.90  | 1–8, $M_n$                            |             | 17  |
|                |              | 30                                    | 63.1                      | 0.28  | 0.37–28, $M_n$                        |             | 18  |
|                | THF          | 30                                    | 73                        | 0.28  | 0.54–4.7, $M_n$                       |             |     |
|                | DMF          | 25                                    | 18.0                      | 0.51  | 1–8, $M_n$                            |             | 17  |
|                | 1N NaOH      |                                       | 16.6                      | 0.48  |                                       |             |     |
|                | (HO) acetone | 30                                    | 8.13                      | 0.5   | 0.69–2.6, $M_n$                       |             | 18  |
|                | PVPDMAEMA    | LiNO <sub>3</sub>                     | 25                        | 1.42  | 0.67                                  | N           | 19  |
| PTAC 50        | 1 M NaCl     | 25                                    | 5.64                      | 0.90  |                                       | O           | 20  |
| PTAC 100       | 1 M NaCl     | 25                                    | 1.55                      | 0.97  |                                       | O           | 20  |
| PTMAC 50       | 1 M NaCl     | 25                                    | 185                       | 0.66  |                                       | O           | 20  |
| PTMAC 100      | 1 M NaCl     | 25                                    | 59.8                      | 0.72  |                                       | O           | 20  |
| PMVEMA         | tris-buffer  | 35                                    | 3.17                      | 0.70  |                                       | P           | 21  |
| PDMDAAC        | 0.1 M NaCl   | 30                                    | 3.5                       | 0.62  |                                       | O           | 22  |

| Polymer                         | Solvent                                                              | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref    |
|---------------------------------|----------------------------------------------------------------------|--------------|---------------------------|-------|---------------------------------------|-------------|--------|
| Poly( <i>N</i> -vinylacetamide) |                                                                      |              |                           |       |                                       |             |        |
|                                 | H <sub>2</sub> O                                                     | 30           | 16                        | 0.52  |                                       | Q           | 22     |
| Polyamic acid                   | DMAc                                                                 | 30           | 2.01                      | 0.79  |                                       | R           | 23     |
| PEG, PEO                        | H <sub>2</sub> O                                                     | 25           | 1.25                      | 0.78  |                                       |             | 24     |
|                                 |                                                                      | 30           | 1.25                      | 0.78  |                                       |             | 13     |
|                                 | LiNO <sub>3</sub>                                                    | 25           | 2.8                       | 0.72  |                                       | N           | 19     |
|                                 |                                                                      | 35           | 3.48                      | 0.70  |                                       | P           | 21     |
| PEG                             | 0.1 N NaNO <sub>3</sub>                                              | 35           | 10.1                      | 0.58  |                                       |             | 25     |
| PEO                             |                                                                      |              | 3.47                      | 0.70  |                                       |             |        |
|                                 | 0.42 N NaOH                                                          | 25           | 2.5                       | 0.66  |                                       |             | 26     |
|                                 | <i>p</i> -dioxane                                                    | 25           | 3.50                      | 0.71  | 0.06–19                               |             | 27     |
|                                 | DMF                                                                  | 25           | 2.40                      | 0.73  | 1–30                                  |             |        |
|                                 |                                                                      | 60           | 5.0                       | 0.653 |                                       | A           | 3      |
|                                 |                                                                      |              | 5.50                      | 0.643 |                                       | A           | 4      |
|                                 | MeOH/H <sub>2</sub> O (1:1)                                          | 25           | 2.21                      | 0.75  |                                       | B           | 5      |
| PEG                             | MeCN/H <sub>2</sub> O (2:8)                                          | 35           | 87.3                      | 0.60  |                                       |             | 25     |
| PEO                             |                                                                      |              | 2.45                      | 0.73  |                                       |             |        |
|                                 | MeOH/H <sub>2</sub> O (2:8)                                          | 25           | 3.4                       | 0.75  |                                       |             | 28     |
| PPG                             | benzene                                                              | 30           | 3.241                     | 0.686 | 1–4                                   |             | 27     |
|                                 | toluene                                                              | 30           | 0.447                     | 0.920 | 0.8–4                                 |             |        |
|                                 | <i>p</i> -dioxane                                                    | 30           | 2.247                     | 0.715 | 1–4                                   |             |        |
|                                 | DMF                                                                  | 30           | 5.284                     | 0.598 | 0.8–4                                 |             |        |
| PVA                             | H <sub>2</sub> O                                                     | 25           | 14.0                      | 0.60  | 10–70                                 |             | 29     |
|                                 | 0.1 N NaNO <sub>3</sub>                                              | 35           | 13.3                      | 0.56  | 13–250                                |             | 30     |
| Cellulose nitrate               | THF                                                                  | 25           | 25.0                      | 1.00  | 95–2300                               |             | 31     |
|                                 |                                                                      |              | 82                        | 1.0   | DP < 1                                | S           | 32, 33 |
|                                 |                                                                      |              | 446                       | 0.76  | DP > 1                                | S           |        |
|                                 |                                                                      |              | 6.44                      | 0.73  |                                       | T           | 34     |
| Nitrocellulose                  | THF                                                                  | 25           | 3.21                      | 0.83  |                                       |             | 35     |
| Methyl cellulose                | H <sub>2</sub> O                                                     | 25           | 10.7                      | 0.61  |                                       | O           | 36     |
| Dextran                         | H <sub>2</sub> O                                                     | 25           | 9.78                      | 0.5   |                                       |             | 24     |
|                                 | 0.1 M NaCl                                                           | 23           | 15.7                      | 0.451 |                                       | U           | 37     |
|                                 |                                                                      | 25           | 13.6                      | 0.47  |                                       |             | 38     |
|                                 | 0.1 M NaNO <sub>3</sub>                                              | 25           | 37                        | 0.40  |                                       | V           | 39     |
|                                 | 0.2 M NaNO <sub>3</sub>                                              |              | 48.5                      | 0.39  |                                       |             | 40     |
|                                 | 0.2 M NaNO <sub>3</sub> +<br>0.025 M KH <sub>2</sub> PO <sub>4</sub> | 30           | 47.6                      | 0.34  |                                       |             | 41     |
|                                 | 0.42 N NaOH                                                          | 25           | 12.0                      | 0.48  |                                       |             | 26     |
|                                 | NaOH (0.005 N)                                                       | 25           | 18.0                      | 0.438 |                                       |             | 42     |
|                                 | (0.0185 N)                                                           |              | 18.5                      | 0.431 |                                       |             |        |
|                                 | (0.097 N)                                                            |              | 15.8                      | 0.451 |                                       |             |        |
|                                 | (0.287 N)                                                            |              | 12.4                      | 0.474 |                                       |             |        |
|                                 | (0.501 N)                                                            |              | 13.2                      | 0.478 |                                       |             |        |
|                                 | (0.671 N)                                                            |              | 14.7                      | 0.465 |                                       |             |        |
|                                 | (0.877 N)                                                            |              | 15.1                      | 0.464 |                                       |             |        |
| Branched dextran                | MeOH/H <sub>2</sub> O                                                |              | 103                       | 0.25  |                                       |             | 43     |
| Pullulan                        | 0.1 M NaCl                                                           | 25           | 1.79                      | 0.67  |                                       |             | 38     |
|                                 | 0.2 M NaNO <sub>3</sub>                                              |              | 4.22                      | 0.64  |                                       |             | 40     |
|                                 | 0.1 M NaNO <sub>3</sub> +<br>0.025 M KH <sub>2</sub> PO <sub>4</sub> | 30           | 1.70                      | 0.72  |                                       |             | 41     |

| Polymer        | Solvent                     | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$   | MW<br>range<br>( $M \times 10^{-3}$ ) | Re-<br>mark | Ref |
|----------------|-----------------------------|--------------|---------------------------|-------|---------------------------------------|-------------|-----|
| Polysaccharide | NaAc buffer                 | 25           | 1.79                      | 0.67  |                                       |             | 44  |
|                | H <sub>2</sub> O            | 25           | 2.23                      | 0.67  |                                       | Q           | 45  |
|                |                             |              | 2.36                      | 0.66  |                                       |             | 46  |
|                | 0.1N NaNO <sub>3</sub>      | 35           | 2.14                      | 0.66  |                                       |             | 25  |
|                | MeCN/H <sub>2</sub> O (2:8) | 35           | 3.72                      | 0.59  |                                       |             |     |
|                | MeOH/H <sub>2</sub> O (2:8) | 25           | 4.4                       | 0.6   |                                       |             | 28  |
|                | H <sub>2</sub> O            | 25           | 6.9                       | 0.6   |                                       |             |     |
| Xanthan        | 0.1M NaNO <sub>3</sub>      | 25           | 0.024                     | 1.10  |                                       | V           | 39  |
| Xylan          | 0.5M NaOH                   | 25           | 2.67                      | 0.73  |                                       |             | 47  |
| NaPSS          | phosphate                   | 25           | 0.60                      | 0.74  |                                       | Q, W        | 45  |
|                |                             | 25           | 0.50                      | 0.80  |                                       | Q, X        | 45  |
|                | NaCl (0.005 N)              | 25           | 0.057                     | 1.061 |                                       |             | 42  |
|                | (0.05 N)                    |              | 0.106                     | 0.935 |                                       |             |     |
|                | (0.20 N)                    |              | 0.209                     | 0.845 |                                       |             |     |
|                | (0.50 N)                    |              | 0.285                     | 0.794 |                                       |             |     |
|                | (1.0 N)                     |              | 0.416                     | 0.741 |                                       |             |     |
|                | 0.1M NaCl                   | 60           | 0.24                      | 0.80  |                                       |             | 48  |
|                | NaOH (0.005 N)              | 25           | 0.038                     | 1.079 |                                       |             | 42  |
|                | (0.0185 N)                  |              | 0.146                     | 0.935 |                                       |             |     |
|                | (0.097 N)                   |              | 0.458                     | 0.805 |                                       |             |     |
|                | (0.287 N)                   |              | 0.281                     | 0.815 |                                       |             |     |
|                | (0.501 N)                   |              | 0.198                     | 0.828 |                                       |             |     |
|                | (0.671 N)                   |              | 0.174                     | 0.830 |                                       |             |     |
|                | (0.877 N)                   |              | 0.110                     | 0.851 |                                       |             |     |
|                | 0.42N NaOH                  | 25           | 0.2                       | 0.82  |                                       |             | 26  |
|                | see Remark                  |              | 1.09                      | 0.74  |                                       | Y           | 10  |
| Chitosan       | see Remark                  | 25           | 19.9                      | 0.59  |                                       | O, Z        | 49  |
| HA             | 0.15M NaCl                  | 25           | 6.54                      | 1.16  |                                       |             | 50  |
|                | see Remark                  |              | 3.6                       | 0.78  |                                       | Z2          | 51  |
| Ch4-S          | 0.2M NaCl                   | 25           | 17                        | 1.01  |                                       |             | 50  |
|                | 0.02M NaCl                  |              | 3.1                       | 0.74  |                                       |             | 52  |
| P(DL)LA        | THF                         | 31.15        | 5.49                      | 0.639 |                                       |             | 53  |
| Proteins       | guanidine                   |              | 3.16                      | 0.66  |                                       | Z3          | 54  |
|                | see Remark                  | 25           | 0.0174                    | 1.09  |                                       | Z4          | 55  |

**Abbreviations.** DMF – *N,N*-dimethylformamide; THF – tetrahydrofuran; TFA – trifluoroacetic acid; NaAc – sodium acetate; phosphate – phosphate buffer; NMP – *N*-methylpyrrolidone; PAN – poly(acrylonitrile); PVP – poly(vinyl pyrrolidone); P2VP – poly(2-vinyl pyridine); NaPA – sodium polyacrylate; phenol (RN) – phenol-formaldehyde resin (random novolak resin); Phenol(HO) – phenol-formaldehyde resin (high-ortho resin); PEG – poly(ethylene glycol); PEO – poly(ethylene oxide); PPG – poly(propylene glycol); PVA – poly(vinyl alcohol); NaPSS – sodium poly(styrene sulfonate); PVPDMAEMA – poly(vinylpyrrolidone-co-dimethylaminoethyl methacrylate); PTAC50 – poly(acrylamide-co-*N,N,N*-trimethylammonium ethyl acrylate chloride); PTAC100 – poly(*N,N,N*-trimethylammonium ethyl acrylate chloride); PTMAC50 – poly(acrylamide-co-*N,N,N*-trimethylammonium ethyl methacrylate chloride); PTMAC100 – poly(*N,N,N*-trimethylammonium ethyl methacrylate chloride); PMVEMA – poly(methyl vinyl ether-co-maleic anhydride); PDMDAAC – poly(dimethyldiallylammonium chloride); HA – hyaluronic acid; Ch4-S – chondroitin 4-sulfate; P(DL)LA – poly(DL-lactic acid).

**Remarks.** A – 0.01 M LiBr included; B – 0.01 M LiNO<sub>3</sub> included; C – 0.1 % TFA + 0.01 N NaNO<sub>3</sub>; D – 0.1 % TFA + 0.20 N NaNO<sub>3</sub>; E – 9.1 % TFA + 0.50 N NaNO<sub>3</sub>; F – use  $K = 0.379 \times 10^{-2}$  and  $a = 0.81$  for PS; G – use  $K = 1.8 \times 10^{-2}$  and  $a = 0.68$  for PS; H – use  $K = 2.88 \times 10^{-2}$  and  $a = 0.60$  for PS; I – 0.1 M sodium formate at pH 3.50; J – 0.5 M sodium formate at pH 3.50; K – 0.1 M sodium formate + 20 % methanol at pH 3.55; L – 80 % (0.5 M Na<sub>2</sub>SO<sub>4</sub> + 0.5 M MeCOOH) + 20 % MeCN; M –  $[\eta] = KM_w^a$ ; N – 0.5 M LiNO<sub>3</sub> (pH 7.0); O –  $[\eta] = KM_w^a$ ; P – 0.1 M tris (hydroxymethyl amino methane) with 0.2 M LiNO<sub>3</sub> adjusted to pH 9.0 with HNO<sub>3</sub>; Q –  $[\eta] = KM_w^a$ ; R – 0.03 M LiBr + 0.03 M H<sub>3</sub>PO<sub>4</sub> + 1 vol % THF in dimethyl acetamide, use  $K = 1.21 \times 10^{-2}$  and  $a = 0.69$  for PS; S –  $[\eta] = K(DP)^a$ , DP – degree of polymerization; T – 0.01 M acetic acid included; U – 0.02 M disodium hydrogen phosphate at pH 7 included; V – 0.2 g/l NaN<sub>3</sub> + 0.1 % ethylene glycol included; W – 0.51 M phosphate buffer (pH 8.0); X – 0.05 M phosphate buffer (pH 8.0); Y – 80 % of 0.03 M Na<sub>2</sub>SO<sub>4</sub> + 20 % MeCN; Z – 0.5 M acetic acid + 0.5 M sodium acetate; Z2 – 0.1 M phosphate + 0.1 M NaCl; Z3 – 6 M guanidinium hydrochloride; Z4 – 0.1 M sodium phosphate + 2 % sodium dodecyl sulfate + 0.02 M dithiothreitol (pH 7.0), proteins (bovine serum albumin, chymotrypsinogen, lysozyme chloride, and cytochrome C).

## References

1. KRANZ D, POHL HU, BAUMANN H (1972) *Angew Makromol Chem* 26:67
2. KENYON AS, MOTTUS EH (1974) *Appl Polym Symp* 25:57
3. MORI S (1983) *Anal Chem* 55:2414
4. AZUMA C, DIAS ML, MANO EB (1986) *Macromol Chem Macromol Symp* 2:169
5. SENAK L, WU CS, MALAWER EG (1987) *J Liq Chromatogr* 10:1127
6. HUGELIN C, DONDOS A (1969) *Macromol Chem* 126:207
7. NAGY DJ, TERWILLIGER DA, LAWREY BD, TIEDGE WF (1989) *International GPC Symposium*, Newton, MA, 637
8. RAND WG, MUKHERJI AK (1982) *J Chromatogr Sci* 20:182
9. PATTON EV, WONNACOTT DM (1987) *J Chromatogr* 389:115
10. KÜHN A, FORSTER S, LÖSCH R, ROMMELFANGER M, ROSENAUER C, SCHMIDT M (1993) *Makromol Chem Rapid Commun* 14:433
11. ISHIGE T, HAMIELEC AE (1973) *J Appl Polym Sci* 17:1479
12. KULICKE WM, KNIIEWSKE R, KLEIN J (1982) *Prog Polym Sci* 8:373
13. SAFIEDDINE AM, HESTER RD (1991) *J Appl Polym Sci* 43:1987
14. KULICKE WM, BÖSE N (1982) *Polym Bull* 7:205
15. MCCARTHY KJ, BURKHARDT CW, PARAZAK DP (1987) *J Appl Polym Sci* 33:1699
16. ROCHAS C, DOMARD A, RINAUDO M (1980) *Eur Polym J* 16:135
17. TOBIASON FL, CHANDLER C, SCHWARZ FE (1972) *Makromolecules* 5:321
18. KAMIDE K, MIYAKAWA Y (1978) *Makromol Chem* 179:359
19. WU CS, SENAK L (1990) *J Liq Chromatogr* 13:851
20. KULICKE WM, JACOBS A (1992) *Makromol Chem Macromol Symp* 61:59
21. WU CS, SENAK L, MALAWER EG (1989) *J Liq Chromatogr* 12:2919
22. DUBIN PL, LEVY IJ (1982) *J Chromatogr* 235:377
23. WALKER CC (1988) *J Polym Sci Part A Polym Chem* 26:1649
24. BAILEY FE, KUCERA JL, IMHOF LG (1958) *J Polym Sci* 32:517
25. NAGY DJ (1990) *J Liq Chromatogr* 13:677
26. CALLEC G, ANDERSON AW, TSAO GT (1984) *J Polym Sci Polym Chem Ed* 22:287
27. ZINBO M, PARSONO JL (1971) *J Chromatogr* 55:55
28. DAWKINS JV, GABBOTT NP, MONTENEGRO AMC (1990) *J Appl Polym Sci Appl Polym Symp* 45:103
29. DIEU HA (1954) *J Polym Sci* 12:417
30. NAGY DJ (1993) *J Liq Chromatogr* 16:3041
31. RUDIN A, HOEGY HW (1972) *J Polym Sci Part A-1*, 10:217
32. MARX-FIGINI M, SOUBELET O (1984) *Polym Bull* 11:281
33. SOUBELET O, PRESTA MA, MARX-FIGINI M (1990) *Angew Makromol Chem* 175:17

34. EREMEEVA TE, BYKOVA TO, GROMOV VS (1990) *J Chromatogr* 522:67
35. BROWN W, WILKSTROM R (1965) *Eur Polym J* 1:1
36. SAKAMOTO N (1987) *Polymer* 28:288
37. MAZSAROFF I, REGNIER FE (1988) *J Chromatogr* 442:15
38. KATO T, TOKUYAMA T, TAKAHASHI A (1983) *J Chromatogr* 256:61
39. TINLAND B, MAZET J, RINAUDO M (1988) *Makromol Chem Rapid Commun* 9:69
40. BAHARY WS, HOGAN MP, JILANI M, ARONSON MP (1995) In: PROVIDER T, BARTH HG, URBAN MW (eds) *Adv Chem Ser* 247, ACS, Washington, p 151
41. BAHARY WS, HOGAN MP (1996) *Int J Polym Anal Charact* 2:121
42. BOSE A, ROLLINGS JE, CARUTHERS JM, OKOS MR, TSAO GT (1982) *J Appl Polym Sci* 27:795
43. CERNY L, McTIERVAN J, STASIW D (1973) *J Polym Sci Symp* 42:1455
44. SOMMERMEYER K, CECHE F, PFITZER E (1988) *Chromatographia* 25:167
45. DUBIN PL, TECKLENBURG MM (1985) *Anal Chem* 57:275
46. KAWAHARA K, OHTA K, MIYAMOTO H, NAKAMURA S (1984) *Carbohydr Polym* 4:335
47. EREMEEVA TE, BYKOVA TO (1993) *J Chromatogr* 639:159
48. MALFAIT T, SLOOTMAEKERS D, CAUWELAERT FV (1990) *J Appl Polym Sci* 39:571
49. YOMOTO C, MIYAZAKI T, OKADA S (1993) *Colloid Polym Sci* 271:76
50. HITTNER DM, COWNAN MK (1987) *J Chromatogr* 402:149
51. LAURENT TC, RYAN M, PIETRUSZKIEWICZ A (1960) *Biochim Biophys Acta* 42:476
52. MATHEWS MB, DORFMAN A (1953) *Arch Biochem Biophys* 42:41
53. VAN DIJK JAPP, SMIT JAM, KOHN FE, FEIJEN J (1983) *J Polym Sci Polym Chem Ed* 21:197
54. NAVE R, WEBER K, POTSCHEKA M (1993) *J Chromatogr A* 654:229
55. CHIKAZUMI N, OHTA T (1991) *J Liq Chromatogr* 14:403

## Appendix III

### Mark-Houwink Parameters for Copolymers

| Polymer    | Solvent                | Temp<br>(°C) | $K \times 10^2$<br>(ml/g) | $a$    | Remark             | Ref |
|------------|------------------------|--------------|---------------------------|--------|--------------------|-----|
| PVC-PVAc   | THF                    | 25           | 6.72                      | 0.611  | 10–13% VAc         | 1   |
| EPM        | toluene                | 30           | 3.051                     | 0.76   | 50% PP, A          | 2   |
| EPDM       | THF                    | 35           | 27.4                      | 0.54   |                    | 3   |
| EPR        | ODCB                   | 135          | 4.43                      | 0.725  |                    | 4   |
|            |                        |              | 2.42                      | 0.707  |                    | 5   |
|            | toluene                | 30           | 3.1                       | 0.72   | 62% PE, B          | 6   |
| E-NORB     | xylene                 | 90           | 10.0                      | 0.535  | 50%                | 7   |
|            | DEB                    | 120          | 4.93                      | 0.589  |                    |     |
| EVA        | THF                    | 20           | 9.7                       | 0.62   | 27–29% EVA, C      | 8   |
| P (S-MMA)  | THF                    | ns           | 0.775                     | 0.76   | alternating        | 9   |
|            | MEK                    |              | 1.15                      | 0.69   |                    |     |
|            | toluene                |              | 1.09                      | 0.73   |                    |     |
|            | <i>n</i> -chlorobutane |              | 1.08                      | 0.70   |                    |     |
|            | THF                    |              | 0.641                     | 0.76   | 1:1 block          |     |
|            | MEK                    |              | 1.466                     | 0.67   |                    |     |
|            | toluene                |              | 1.318                     | 0.69   |                    |     |
|            | <i>n</i> -chlorobutane |              | 4.446                     | 0.55   |                    |     |
|            | toluene                | ns           | 1.14                      | 0.70   | statistical, 30% S | 10  |
|            | <i>n</i> -chlorobutane |              | 2.65                      | 0.60   |                    |     |
|            | toluene                |              | 1.32                      | 0.71   | 57% S              |     |
|            | <i>n</i> -chlorobutane |              | 2.49                      | 0.63   |                    |     |
|            | toluene                |              | 0.832                     | 0.75   | 71% S              |     |
|            | <i>n</i> -chlorobutane |              | 1.76                      | 0.67   |                    |     |
| P (S-MAH)  | THF                    | 25           | 3.98                      | 0.595  | MAH 5– 50mol%, B   | 11  |
| P (S-AN)   | DMF                    | 20           | 1.45                      | 0.71   | 12% S, D           | 12  |
|            |                        |              | 1.80                      | 0.71   | 25% S              |     |
|            |                        |              | 2.65                      | 0.72   | 52% S              |     |
|            |                        | 25           | 1.86                      | 0.690  | 15–82% AN, E       | 13  |
|            | THF                    |              | 1.98                      | 0.681  | 0–35% AN, E        |     |
|            | MEK                    |              | 3.30                      | 0.619  | 10–48% AN, E       |     |
| P (AN-EVE) | DMF                    | 55           | 0.142                     | 0.9636 | 18% EVE, F         | 14  |
| P (SA-AA)  | 0.5M NaCl              | 25           | 0.81                      | 0.789  | 10% SA, G, pH 9    | 15  |
|            | 1M NaCl                |              | 0.307                     | 0.860  |                    |     |
|            | 0.5M NaCl              |              | 1.09                      | 0.775  | 20% SA             |     |
|            | 1M NaCl                |              | 1.307                     | 0.759  |                    |     |
|            | 0.5M NaCl              |              | 5.78                      | 0.655  | 50% SA             |     |
|            | 1M NaCl                |              | 2.98                      | 0.695  |                    |     |
|            | 0.5M NaCl              |              | 0.638                     | 0.844  | 70% SA             |     |
|            | 1M NaCl                |              | 0.026                     | 1.072  |                    |     |

**Abbreviations.** THF – tetrahydrofuran; ODCB – *o*-dichlorobenzene; DEB – diethylbenzene; MEK – methyl ethyl ketone; ns – not specified; PVC – polyvinyl chloride; PVAc – polyvinyl acetate; EPM – ethylene-propylene copolymer; EPDM – terpolymer of ethylene, propylene, diene monomers; EPR – ethylene-propylene rubber; E-NORB – ethylene-norbornene (50%); EVA – ethylene-vinyl acetate copolymer; S – polystyrene (PS); MMA – polymethyl methacrylate (PMMA); MAH – maleic anhydride; AN – polyacrylonitrile (PAN); EVE – ethyl vinyl ether; SA – sodium acrylate; AA – acrylamide.

**Remarks.** A – use  $K = 1.112 \times 10^{-2}$  and  $a = 0.723$  for PS; B –  $[\eta] = KM_w^a$ ; C – use  $K = 1.55 \times 10^{-2}$  and  $a = 0.692$  for PS; D – use  $K = 2.40 \times 10^{-2}$  and  $a = 0.63$  for PS and  $K = 3.25 \times 10^{-2}$  and  $a = 0.725$  for PAN; E –  $[\eta] = K(W_{AN} + 1.041)^a M_w^a$ ,  $W_{AN}$ : weight fraction of AN unit; F – 0.05 M LiBr included, use  $K = 2.2 \times 10^{-2}$  and  $a = 0.615$  for PS; G –  $[\eta] = KM_w^a$ ; H – use  $K = 1.17 \times 10^{-2}$  and  $a = 0.717$  for PS.

## References

1. JANČA J, MRKVIČKOVÁ L, KOLÍNSKY M (1978) J Appl Polym Sci 22:2661
2. SMITH WV (1974) J Appl Polym Sci 18:3685
3. CHIANTORE O, CINQUINA P, GUAITA M (1994) Eur Polym J 30:1043
4. GIANOTTI G, CICUTA A, ROMANINI D (1980) Polymer 21:1087
5. ATKINSON CML, DIETZ R (1976) Makromol Chem 177:213
6. DE CHIRICO A, ARRIGHETTI S, BRUZZONE M (1981) Polymer 22:529
7. BRAUER E, WIEGLEB H, HELMSTEDT M (1986) Polym Bull 15:551
8. ECHARRI J, IRUIN JJ, GUZMAN GM, AMSORENA J (1979) Makromol Chem 180:2749
9. KOTAKA T, TANAKA T, ONHUMI H, MURAKAMI Y, INAGAKI H (1970) Polym J 1:245
10. GOLDWASSER JM, RUDIN A (1983) J Liq Chromatogr 6:2433
11. CHOW CD (1976) J Appl Polym Sci 20:1619
12. KRANZ D, POHL HU, BAUMANN H (1972) Angew Makromol Chem 26:67
13. MENDELSON RA (1987) In: PROVIDER T (ed) ACS Symposium Series No 352, ACS, Washington, p 263
14. KENYON AS, MOTTUS EH (1974) Appl Polym Symp 25:57
15. MCCARTHY KJ, BURKHARDT CW, PARAZAK DP (1987) J Appl Polym Sci 33:1699



## Appendix IV

### Specific Refractive Index Increments

| Sample      | Solvent                   | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | $A_2 (\times 10^3)$ | Ref, Remark |
|-------------|---------------------------|------------------|--------------|-----------|---------------------|-------------|
| Polystyrene | toluene                   | 632.8            | 0.110        | 23        |                     | 1           |
|             |                           | 632.8            | 0.108        | 25        |                     | 2, A        |
|             |                           | 632.8            | 0.108        |           |                     | 3           |
|             |                           | 632.8            | 0.107        |           |                     | 4           |
|             |                           | 632.8            | 0.109        | 22        | 0.5–0.3             | 5           |
|             |                           | 632.8            | 0.111        | 20        |                     | 6           |
|             |                           | 670              | 0.108        |           |                     | 7           |
|             |                           | 589.3            | 0.107        |           |                     | 8, C        |
|             | benzene                   | 546.0            | 0.104        |           |                     | 9           |
|             |                           | 546.0            | 0.1064       |           |                     | 9           |
|             |                           | 546.0            | 0.109        | 25        |                     | 10          |
|             |                           | 436              | 0.112        | 25        |                     | 10          |
|             | CCl <sub>4</sub>          | 546.0            | 0.146        |           |                     | 9           |
|             |                           | 488              | 0.199        |           |                     | 10          |
|             | tetrahydrofuran (THF)     | 632.8            | 0.190        |           |                     | 3           |
|             |                           | 632.8            | 0.193        | 20        |                     | 6           |
|             |                           | 632.8            | 0.1845       | 25        |                     | 11          |
|             |                           | 632.8            | 0.184        | 30        |                     | 12          |
|             |                           | 632.8            | 0.186        |           |                     | 4           |
|             |                           | 632.8            | 0.184        |           |                     | 13          |
|             |                           | 632.8            | 0.188        | 22        | 0.47–0.3            | 5           |
|             |                           | 632.8            | 0.192        | 25        |                     | 14          |
|             |                           | 632.8            | 0.184        | 40        |                     | 15          |
|             |                           | 670              | 0.180        | 25        |                     | 11          |
|             |                           | 670              | 0.1805       | 30        |                     | 16, D       |
|             | chloroform                | 632.8            | 0.155        |           |                     | 3           |
|             |                           | 632.8            | 0.1608       |           |                     | 17          |
|             |                           | 546.0            | 0.169        | 25        |                     | 10          |
|             |                           | 436              | 0.161        | 25        |                     | 10          |
|             | acetone                   | 670              | 0.224        |           |                     | 7           |
|             | methyl ethyl ketone (MEK) | 632.8            | 0.212        | 22        | 0.125–0.06          | 5           |
|             |                           | 632.8            | 0.214        |           |                     | 7           |
|             |                           | 670              | 0.213        |           |                     | 7           |
|             |                           | 589.3            | 0.212        |           |                     | 8, C        |
|             |                           | 546.0            | 0.220        |           |                     | 9           |
|             | tetrahydronaphthalene     | 670              | 0.0615       |           |                     | 7           |

| Sample                              | Solvent                   | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | $A_2 (\times 10^3)$ | Ref, Remark |
|-------------------------------------|---------------------------|------------------|--------------|-----------|---------------------|-------------|
| Polystyrene                         | MC/DCAA (8:2)             | 632.8            | 0.156        |           |                     | 18, E       |
|                                     | ethyl benzoate            | 632.8            | 0.103        | 15        |                     | 19          |
|                                     | dimethyl-                 | 632.8            | 0.165        | 20        |                     | 7           |
|                                     | formamide (DMF)           | 632.8            | 0.159        |           |                     | 20, F       |
|                                     | tetrachloroethylene (TCE) | 632.8            | 0.0934       | 25        |                     | 21          |
|                                     | <i>p</i> -dioxane         | 546.0            | 0.152        | 35        |                     | 22          |
|                                     | (star)                    | 546.0            | 0.168        | 35        |                     | 22          |
|                                     | cyclohexane               | 546.0            | 0.170        | 35        |                     | 22          |
|                                     | (star)                    | 546.0            | 0.168        | 35        |                     | 22          |
|                                     |                           |                  |              |           |                     |             |
| Poly(dimethylsiloxane)              | TCE                       | 632.8            | -0.0932      | 25        |                     | 21          |
|                                     | toluene                   | 632.8            | -0.0913      | 26        |                     | 20          |
|                                     |                           | 632.8            | -0.0648      | 26        |                     | 20          |
| Poly(styrene-divinylbenzene)        | DMF/LiBr                  | 546              | 0.170        | 30        |                     | 23          |
|                                     | THF                       | 546.1            | 0.1963       | 30        |                     | 24          |
| Poly(styrene-methyl methacrylate)   | ethyl benzoate            | 632.8            | 0.0398       | 15        |                     | 19, G       |
| Poly(methyl methacrylate)           | THF                       | 632.8            | 0.083        | 25        | 0.2                 | 25          |
|                                     |                           | 632.8            | 0.084        |           |                     | 26, H       |
|                                     |                           | 632.8            | 0.084        |           |                     | 13          |
|                                     |                           | 632.8            | 0.084        |           |                     | 14          |
|                                     |                           | 632.8            | 0.084        |           |                     | 27          |
|                                     |                           | 632.8            | 0.09         |           |                     | 28          |
|                                     |                           | 589              | 0.0865       | 23        |                     | 29          |
|                                     |                           | 546              | 0.0871       | 25        |                     | 10          |
|                                     | HFIP                      | 436              | 0.0887       | 25        |                     | 10          |
|                                     |                           | 632.8            | 0.190        | 25        |                     | 30, J       |
|                                     |                           | 632.8            | 0.190        |           |                     | 27          |
|                                     |                           | 546              | 0.0157       | 25        |                     | 10          |
|                                     | toluene                   | 488              | 0.010        | 18        |                     | 19          |
|                                     |                           | 439              | 0.0094       | 25        |                     | 10          |
|                                     |                           | 632.8            | 0.053        | 25        |                     | 31          |
|                                     |                           | MEK              | 546          | 0.1140    |                     | 9           |
|                                     |                           | 546              | 0.1173       | 25        |                     | 10          |
|                                     |                           | 436              | 0.100        | 25        |                     | 10          |
|                                     |                           | 546.0            | -0.010       |           |                     | 9           |
|                                     |                           | 546.0            | -0.011       | 25        |                     | 10          |
|                                     | benzene                   | 436              | -0.0039      | 25        |                     | 10          |
|                                     |                           | CCl <sub>4</sub> | 546.0        | 0.023     |                     | 9           |
|                                     |                           | acetone          | 546.0        | 0.1340    |                     | 9           |
|                                     |                           |                  |              |           |                     |             |
| Poly(ethyl methacrylate)            | MEK                       | 546.0            | 0.104        | 23        |                     | 10          |
|                                     | ethyl acetate             | 436              | 0.109        | 25        |                     | 10          |
| Poly( <i>n</i> -butyl methacrylate) | MEK                       | 546.0            | 0.102        | 25        |                     | 10          |
|                                     |                           | 436              | 0.103        | 25        |                     | 10          |
|                                     | benzene                   | 546.0            | -0.014       | 25        |                     | 10          |
|                                     |                           | 436              | -0.023       | 25        |                     | 10          |

| Sample                             | Solvent        | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|------------------------------------|----------------|------------------|--------------|-----------|-------------------------------------|-------------|
| Poly(methacrylic acid)             | ethanol        | 436              | 0.154        | 25        |                                     | 10          |
|                                    | methanol       | 546              | 0.134        | 25        |                                     | 10          |
| Poly(methyl acrylate)              | benzene        | 436              | -0.019       | 30        |                                     | 10          |
|                                    | acetonitrile   | 546.0            | 0.118        | 25        |                                     | 10          |
|                                    |                | 436              | 0.120        | 25        |                                     | 10          |
| Poly(ethyl acrylate)               | acetone        | 546.0            | 0.107        | 23        |                                     | 10          |
|                                    |                | 436              | 0.109        | 23        |                                     | 10          |
|                                    | chloroform     | 546.0            | 0.0363       | 25        |                                     | 10          |
|                                    | MEK            | 546.0            | 0.0856       | 20        |                                     | 10          |
|                                    |                | 436              | 0.0870       | 20        |                                     | 10          |
| Poly( <i>n</i> -butyl acrylate)    | acetone        | 436              | 0.0870       | 20        |                                     | 10          |
|                                    | benzene        | 546              | -0.0292      | 30        |                                     | 10          |
|                                    | THF            | 546              | 0.0651       | 30        |                                     | 10          |
|                                    |                | 632.8            | 0.067        |           |                                     | 6           |
|                                    | toluene        | 546              | -0.0239      | 30        |                                     | 10          |
|                                    |                | 632.8            | -0.024       | 20        |                                     | 6           |
|                                    | MEK            | 632.8            | 0.097        | 20        |                                     | 6           |
|                                    | DMF            | 632.8            | 0.016        | 20        |                                     | 6           |
|                                    |                |                  |              |           |                                     |             |
| Poly( <i>tert</i> -butyl acrylate) | MEK            | 546              | 0.0818       | 25        |                                     | 10          |
|                                    | propyl alcohol | 546              | 0.074        |           |                                     | 32          |
| Poly(vinyl chloride)               | THF            | 632.8            | 0.101        |           |                                     | 20          |
|                                    |                | 589.0            | 0.106        | 23        |                                     | 29          |
|                                    |                | 589.0            | 0.115        |           |                                     | 29          |
|                                    |                | 589.0            | 0.115        | 25        |                                     | 10          |
|                                    |                | 578              | 0.1010       | 16.5      |                                     | 10          |
|                                    |                | 546.0            | 0.1017       | 16.5      |                                     | 10          |
|                                    |                | 546.0            | 0.116        | 25        |                                     | 10          |
|                                    |                | 435.8            | 0.115        | 20        |                                     | 33          |
|                                    |                | 435.8            | 0.1124       | 20        |                                     | 34          |
|                                    |                | 435.8            | 0.1074       | 25        |                                     | 35          |
|                                    |                | 435.8            | 0.104        | 16.5      |                                     | 10          |
|                                    |                | 435.8            | 0.119        | 25        |                                     | 10          |
|                                    | cyclohexanone  | 632.8            | 0.0650       |           |                                     | 20          |
|                                    |                | 589.0            | 0.076        | 25        |                                     | 10          |
|                                    |                | 546.0            | 0.077        | 25        |                                     | 10          |
|                                    |                | 435.8            | 0.079        | 25        |                                     | 10          |
| Polyisoprene                       | toluene        | 632.8            | 0.031        |           |                                     | 3           |
|                                    | THF            | 632.8            | 0.127        |           |                                     | 3           |
|                                    |                | 589.0            | 0.156        |           |                                     | 29          |
|                                    | chloroform     | 632.8            | 0.093        |           |                                     | 3           |
|                                    | benzene        | 546.0            | 0.0194       | 20        |                                     | 10          |
| ( <i>cis</i> -1,4)                 |                | 436              | 0.0143       | 20        |                                     | 10          |
| ( <i>trans</i> -1,4)               | cyclohexane    | 436              | 0.117        |           |                                     | 10          |
| Polybutadiene                      | toluene        | 632.8            | 0.032        |           |                                     | 3           |
|                                    | THF            | 632.8            | 0.132        |           |                                     | 3           |
|                                    |                | 632.8            | 0.1295       | 26        |                                     | 20, K       |
|                                    |                | 589.0            | 0.132        | 40        |                                     | 29          |
|                                    | chloroform     | 632.8            | 0.094        |           |                                     | 3           |
|                                    |                | 546.0            | 0.083        |           |                                     | 10          |

| Sample                                                                | Solvent                | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|-----------------------------------------------------------------------|------------------------|------------------|--------------|-----------|-------------------------------------|-------------|
| Polybutadiene<br>( <i>cis</i> -1,4)<br>( <i>trans</i> -1,4)<br>(1,2-) | cyclohexane            | 632.8            | 0.1151       | 30        |                                     | 36          |
|                                                                       |                        | 632.8            | 0.1084       | 30        |                                     | 36          |
|                                                                       |                        | 632.8            | 0.0873       | 30        |                                     | 36          |
|                                                                       |                        | 632.8            | 0.111        | 30        |                                     | 36, L       |
|                                                                       |                        | 546              | 0.118        |           |                                     | 10          |
|                                                                       |                        | 546              | 0.1174       | 25        |                                     | 9           |
|                                                                       |                        | 436              | 0.126        |           |                                     | 10          |
|                                                                       | benzene                | 436              | 0.1345       | 25        |                                     | 9           |
|                                                                       |                        | 546              | 0.0086       | 20        |                                     | 10          |
|                                                                       |                        | 546              | 0.0099       | 25        |                                     | 9           |
|                                                                       |                        | 436              | 0.0117       | 20        |                                     | 10          |
|                                                                       | hexane                 | 436              | 0.0160       | 25        |                                     | 9           |
|                                                                       |                        | 546              | 0.1523       | 25        |                                     | 9           |
|                                                                       |                        | 436              | 0.2181       | 25        |                                     | 9           |
| Polychloroprene                                                       | THF                    | 632.8            | 0.1274       | 25        |                                     | 11          |
|                                                                       | MEK                    | 546              | 0.156        | 25        |                                     | 10          |
|                                                                       |                        | 436              | 0.162        | 25        |                                     | 10          |
| Polyisobutylene                                                       | 1-chloronaphthalene    | 632.8            | -0.122       | 135       |                                     | 37          |
|                                                                       |                        | 578              | -0.128       | 135       |                                     | 37          |
|                                                                       |                        | 546              | -0.131       | 135       |                                     | 37          |
|                                                                       |                        | 436              | -0.146       | 135       |                                     | 37          |
|                                                                       | trichlorobenzene (TCB) | 632.8            | -0.051       | 135       |                                     | 37          |
|                                                                       |                        | 578              | -0.053       | 135       |                                     | 37          |
|                                                                       |                        | 546              | -0.054       | 135       |                                     | 37          |
|                                                                       |                        | 436              | -0.066       | 135       |                                     | 37          |
|                                                                       | THF                    | 632.8            | 0.1026       | 23        |                                     | 20, M       |
|                                                                       |                        | 632.8            | 0.125        | 25        | 0.35                                | 38          |
|                                                                       |                        | 546.1            | 0.114        | 25        |                                     | 39          |
|                                                                       | toluene                | 632.8            | 0.106        | 25        |                                     | 2, B        |
| Poly(1-butene)                                                        | 1-chloronaphthalene    | 632.8            | -0.177       | 135       |                                     | 37          |
|                                                                       |                        | 578              | -0.183       | 135       |                                     | 37          |
|                                                                       |                        | 546              | -0.187       | 135       |                                     | 37          |
|                                                                       |                        | 436              | -0.206       | 135       |                                     | 37          |
|                                                                       | TCB                    | 632.8            | -0.103       | 135       |                                     | 37          |
|                                                                       |                        | 578              | -0.105       | 135       |                                     | 37          |
|                                                                       |                        | 546              | -0.108       | 135       |                                     | 37          |
|                                                                       |                        | 436              | -0.120       | 135       |                                     | 37          |
|                                                                       | <i>n</i> -nonane       | 436              | -0.108       | 35        |                                     | 10          |
|                                                                       |                        |                  |              |           |                                     |             |
| Polyethylene                                                          | TCB                    | 632.8            | -0.097       | 135       |                                     | 20          |
|                                                                       |                        | 632.8            | -0.104       | 135       |                                     | 12          |
|                                                                       |                        | 632.8            | -0.098       | 145       |                                     | 12          |
|                                                                       |                        | 632.8            | -0.098       | 145       |                                     | 40          |
|                                                                       |                        | 632.8            | -0.091       | 145       |                                     | 41          |
|                                                                       |                        | 546              | -0.106       | 135       |                                     | 10          |
|                                                                       | 1-chloronaphthalene    | 589.3            | -0.1063      | 135       |                                     | 8, C        |
|                                                                       |                        | 589.3            | -0.1085      |           |                                     | 8, C        |
|                                                                       |                        | 632.8            | -0.177       | 145       |                                     | 20          |
|                                                                       |                        |                  |              |           |                                     |             |

| Sample          | Solvent                          | Wave-length (nm)    | $dn/dc$ (ml/g) | Temp (°C) | $A_2$ ( $\times 10^3$ ) | Ref, Remark |
|-----------------|----------------------------------|---------------------|----------------|-----------|-------------------------|-------------|
| Polyethylene    | decane                           | 589.3               | -0.1883        |           |                         | 8           |
|                 |                                  | 546.0               | -0.191         | 135       |                         | 42          |
|                 |                                  | 577                 | 0.1200         | 125       |                         | 43          |
|                 |                                  | 546                 | 0.1213         | 125       |                         | 43          |
|                 |                                  | 436                 | 0.1263         | 125       |                         | 43          |
|                 | <i>o</i> -dichlorobenzene (ODCB) | 404                 | 0.1233         | 125       |                         | 43          |
|                 |                                  | 632.8               | -0.056         | 135       |                         | 20          |
|                 |                                  | 589                 | -0.090         | 135       |                         | 29          |
|                 | 1-cyclonaphthalene               | 546                 | -0.078         | 120       |                         | 10          |
|                 |                                  | 546                 | -0.214         | 125       |                         | 10          |
|                 |                                  | 436                 | -0.245         | 125       |                         | 10          |
| HDPE            | TCB                              | 632.8               | -0.112         | 145       | 2.00                    | 44          |
|                 |                                  | 632.8               | -0.107         | 135       |                         | 37          |
|                 |                                  | 578                 | -0.108         | 135       |                         | 37          |
|                 |                                  | 546                 | -0.110         | 135       |                         | 37          |
|                 |                                  | 436                 | -0.125         | 135       |                         | 37          |
|                 | (SRM 1475) ODCB                  | 632.8               | -0.104         | 135       | 1.90                    | 45          |
|                 |                                  | 632.8               | -0.056         | 145       | 1.41                    | 44          |
|                 |                                  | 589                 | -0.069         | 135       |                         | 29          |
|                 | (SRM 1475) 1-chloronaphthalene   | 632.8               | -0.056         | 135       |                         | 45          |
|                 |                                  | 632.8               | -0.189         | 145       | 0.68                    | 44          |
|                 |                                  | 632.8               | -0.183         | 135       |                         | 37          |
|                 |                                  | 578                 | -0.188         | 135       |                         | 37          |
|                 |                                  | 546                 | -0.192         | 135       |                         | 37          |
|                 | (SRM 1475) diphenylmethane       | 436                 | -0.215         | 135       |                         | 37          |
|                 |                                  | 632.8               | -0.177         | 145       |                         | 45          |
|                 |                                  | (SRM 1475) 589.3    | -0.1932        | 135       |                         | 8           |
|                 |                                  | 546                 | -0.125         | 142       |                         | 46          |
|                 | LDPE TCB                         | 632.8               | -0.108         | 145       | 3.05                    | 44          |
|                 |                                  | 632.8               | -0.102         | 135       |                         | 37          |
|                 |                                  | 578                 | -0.103         | 135       |                         | 37          |
|                 |                                  | 546                 | -0.105         | 135       |                         | 37          |
|                 |                                  | 436                 | -0.117         | 135       |                         | 37          |
| (SRM 1476) LDPE | TCB                              | 632.8               | -0.091         | 135       | 4.3                     | 45          |
|                 |                                  | 632.8               | -0.091         | 140       | 0.42                    | 47          |
|                 | ODCB                             | 632.8               | -0.051         | 145       | 1.71                    | 44          |
|                 |                                  | 589.0               | -0.104         | 138       |                         | 29          |
|                 |                                  | 632.8               | -0.185         | 145       | 0.87                    | 44          |
|                 | 1-chloronaphthalene              | 632.8               | -0.178         | 135       |                         | 37          |
|                 |                                  | 578                 | -0.184         | 135       |                         | 37          |
|                 |                                  | 546                 | -0.189         | 135       |                         | 37          |
|                 |                                  | 546                 | -0.195         | 125       |                         | 48          |
|                 |                                  | 436                 | -0.212         | 135       |                         | 37          |
|                 | (SRM 1476) TCB                   | 589.3               | -0.1916        | 135       |                         | 8           |
| Linear-LDPE     | TCB                              | 632.8               | -0.106         | 145       | 1.50                    | 44          |
|                 |                                  | 632.8               | -0.106         | 145       |                         | 49, N       |
|                 |                                  | 632.8               | -0.103         | 145       |                         | 49, O       |
|                 | ODCB                             | 632.8               | -0.048         | 145       | 1.05                    | 44          |
|                 |                                  | 1-chloronaphthalene | 632.8          | -0.180    | 145                     | 0.79        |
|                 |                                  |                     |                |           |                         | 44          |

| Sample                             | Solvent                | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|------------------------------------|------------------------|------------------|--------------|-----------|-------------------------------------|-------------|
| Polypropylene                      | 1-chloronaphthalene    | 546              | -0.227       | 125       |                                     | 10          |
|                                    |                        | 546              | -0.191       | 125       |                                     | 27          |
|                                    |                        | 436              | -0.228       | 125       |                                     | 10          |
|                                    |                        | 632.8            | -0.173       | 135       |                                     | 37          |
|                                    |                        | 578              | -0.180       | 135       |                                     | 37          |
|                                    |                        | 546              | -0.184       | 135       |                                     | 37          |
|                                    |                        | 436              | -0.205       | 135       |                                     | 37          |
|                                    |                        | 632.8            | -0.177       | 135       |                                     | 37          |
|                                    |                        | 578              | -0.183       | 135       |                                     | 37          |
|                                    |                        | 546              | -0.188       | 135       |                                     | 37          |
|                                    | TCB                    | 436              | -0.211       | 135       |                                     | 37          |
|                                    |                        | 632.8            | -0.095       | 135       |                                     | 50          |
|                                    |                        | 632.8            | -0.093       | 145       |                                     | 49          |
|                                    |                        | 632.8            | -0.091       | 145       |                                     | 20          |
|                                    |                        | 632.8            | -0.092       | 145       |                                     | 51          |
|                                    |                        | 632.8            | -0.094       | 145       |                                     | 51          |
|                                    |                        | 632.8            | -0.102       | 135       |                                     | 37          |
|                                    |                        | 578              | -0.105       | 135       |                                     | 37          |
|                                    |                        | 546              | -0.107       | 135       |                                     | 37          |
|                                    |                        | 436              | -0.121       | 135       |                                     | 37          |
|                                    |                        | 632.8            | -0.105       | 135       |                                     | 37          |
|                                    |                        | 578              | -0.109       | 135       |                                     | 37          |
|                                    |                        | 546              | -0.111       | 135       |                                     | 37          |
|                                    |                        | 436              | -0.123       | 135       |                                     | 37          |
| Polyoxymethylene                   | DMF                    | 632.8            | -0.0505      | 135       |                                     | 52          |
| Polyethylene terephthalate         | HFIP                   | 632.8            | 0.257        |           | 2.9                                 | 53          |
|                                    | HFIP/PFP (5:5)         | 632.8            | 0.242        |           |                                     | 53, P       |
|                                    | MC/DCAA (8:2)          | 632.8            | 0.148        | RT        |                                     | 18, E       |
|                                    | DCAA                   | 589              | 0.106        |           |                                     | 54, Q       |
|                                    | <i>m</i> -cresol       | 589.0            | -0.073       | 25        |                                     | 29          |
|                                    | <i>o</i> -chlorophenol | 589.0            | -0.070       | 25        |                                     | 29          |
| Polyamide 6                        | HFIP                   | 632.8            | 0.2375       | 25        |                                     | 55          |
|                                    | MC/DCAA (8:2)          | 632.8            | 0.180        | RT        |                                     | 56          |
|                                    | <i>o</i> -chlorophenol | 589.0            | -0.012       | 90        |                                     | 29          |
| Polyamide 11                       | MC/DCAA (8:2)          | 632.8            | 0.133        | RT        |                                     | 56          |
|                                    | HFIP                   | 546              | 0.25         | 30        | 7.8-5.6                             | 57          |
| Polyamide 12                       | MC/DCAA (8:2)          | 632.8            | 0.133        | RT        |                                     | 56          |
|                                    | HFIP                   | 632.8            | 0.2212       |           |                                     | 58          |
|                                    |                        | 546              | 0.25         | 30        |                                     | 59          |
| Polyamide 6, 6<br>(46 K)<br>(32 K) | MC/DCAA (8:2)          | 632.8            | 0.182        | RT        |                                     | 56          |
|                                    |                        |                  | 0.188        | RT        |                                     | 56          |
|                                    |                        |                  | 0.194        | RT        |                                     | 56          |
|                                    | HFIP                   | 632.8            | 0.241        | 25        | 4.8                                 | 60          |
|                                    | formic acid            | 632.8            | 0.136        | 25        |                                     | 60          |
|                                    | <i>m</i> -cresol       | 589.0            | -0.016       | 130       |                                     | 29          |
| Polyamide 6, 9                     | MC/DCAA (8:2)          | 632.8            | 0.163        | RT        |                                     | 56          |

| Sample                                     | Solvent                              | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|--------------------------------------------|--------------------------------------|------------------|--------------|-----------|-------------------------------------|-------------|
| Polyamide 6, 10                            | MC/DCAA (8:2)                        | 632.8            | 0.163        | RT        |                                     | 56          |
| Polyamide 6, 12                            | MC/DCAA (8:2)                        | 632.8            | 0.149        | RT        |                                     | 56          |
| Polyamide 6T                               | MC/DCAA (8:2)                        | 632.8            | 0.201        | RT        |                                     | 56          |
| Silk (Silkworm)                            | HFIP                                 | 632.8            | 0.236        |           |                                     | 61          |
| (Spider)                                   | HFIP                                 | 632.8            | 0.235        |           |                                     | 61          |
| Polyacrylonitrile                          | DMF                                  | 632.8            | 0.092        | 25        |                                     | 31          |
|                                            |                                      | 546              | 0.082        | 20        |                                     | 10          |
|                                            |                                      | 436              | 0.087        | 20        |                                     | 10          |
|                                            | dimethylsulfoxide                    | 546              | 0.031        | 25        |                                     | 10          |
|                                            |                                      | 436              | 0.029        | 25        |                                     | 10          |
|                                            | dimethylacetamide                    | 546              | 0.0769       | 25        |                                     | 10          |
|                                            |                                      | 436              | 0.0767       | 25        |                                     | 10          |
|                                            | THF                                  | 632.8            | 0.1546       | 23        |                                     | 20          |
| Poly(acrylonitrile-styrene)<br>(28.5/71.5) |                                      |                  |              |           |                                     |             |
| Polyurethane                               | THF                                  | 632.8            | 0.0863       | 25        |                                     | 62          |
|                                            |                                      | 632.8            | 0.1469       | 23        |                                     | 20, R       |
| Poly(methyl vinyl ether)                   | cyclohexane                          | 632.8            | 0.050        | 50        |                                     | 63          |
| PMVEMA                                     | 0.1 M tris                           | 632.8            | 0.227        |           |                                     | 64, S       |
| Poly(phenyl acetylene)                     | THF                                  | 632.8            | 0.2864       | 25        |                                     | 65          |
| Epoxy resin                                | THF                                  | 632.8            | 0.187        | 23        |                                     | 20          |
| Phenoxy resin                              | THF                                  | 632.8            | 0.1773       |           |                                     | 26          |
| Polycarbonate                              | THF                                  | 632.8            | 0.177        |           |                                     | 20          |
|                                            |                                      | 632.8            | 0.1855       |           |                                     | 20          |
| Polyphosphazene I<br>II<br>III<br>IV       | THF                                  | 632.8            | 0.160        | 30        |                                     | 66, T       |
|                                            |                                      | 632.8            | 0.145        | 30        |                                     | 66, T       |
|                                            |                                      | 632.8            | -0.029       | 30        |                                     | 66, T       |
|                                            |                                      | 632.8            | -0.023       | 25        | 1.0                                 | 67, T       |
|                                            | acetone<br>cyclohexanone             | 632.8            | 0.019        | 25        | 5.1                                 | 67, T       |
|                                            |                                      | 632.8            | -0.053       | 40        | 6.7                                 | 67, T       |
|                                            |                                      |                  |              |           |                                     |             |
|                                            |                                      |                  |              |           |                                     |             |
| Poly(ether sulfone)                        | DMF                                  | 632.8            | 0.188        | 25        |                                     | 4           |
| Polyimide                                  | DMF                                  | 632.8            | 0.208        | 25        |                                     | 4           |
|                                            | N-methylpyrrolidone                  | 632.8            | 0.132        |           |                                     | 68, U       |
| PEEK                                       |                                      |                  |              |           |                                     |             |
| (MW 14300-22600)                           | conc. H <sub>2</sub> SO <sub>4</sub> | 632.8            | 0.376        |           | 4.3-2.0                             | 69, V       |
| (MW 28300-56500)                           |                                      | 632.8            | 0.419        |           | 1.8-2.7                             | 69, V       |
| Poly(phenylene sulfide)                    |                                      |                  |              |           |                                     |             |
|                                            | 1-chloronaphthalene                  | 632.8            | 0.167        | 220       |                                     | 70          |
| Poly(vinyl butyral)                        | HFIP                                 | 632.8            | 0.189        | 25        |                                     | 30, J       |
|                                            | THF                                  | 632.8            | 0.089        |           |                                     | 30, J       |

| Sample                                                                                  | Solvent                                | Wave-length (nm)         | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |       |       |
|-----------------------------------------------------------------------------------------|----------------------------------------|--------------------------|--------------|-----------|-------------------------------------|-------------|-------|-------|
| Poly(2-vinylpyridine)                                                                   | benzene                                | 546                      | 0.075        | 25        |                                     | 71          |       |       |
| Poly(4-vinylpyridine)                                                                   | DMF                                    | 436                      | 0.160        | 25        |                                     | 10          |       |       |
|                                                                                         | chloroform                             | 436                      | 0.150        | 25        |                                     | 10          |       |       |
| Poly(vinylpyrrolidone)                                                                  | DMF                                    | 632.8                    | 0.095        | 23        |                                     | 20          |       |       |
|                                                                                         | chloroform                             | 546                      | 0.108        |           |                                     | 10          |       |       |
|                                                                                         |                                        |                          | 436          | 0.108     |                                     |             | 10    |       |
|                                                                                         |                                        |                          | 546          | 0.185     |                                     |             | 10    |       |
|                                                                                         | water                                  | 546                      | 0.185        |           |                                     | 10          |       |       |
|                                                                                         |                                        |                          | 436          | 0.185     |                                     |             | 10    |       |
| 6 FDA/p-PDA                                                                             | N-methylpyrrolidone                    | 632.8                    | 0.107        |           |                                     | 72, W       |       |       |
| 6 FDA/MDA                                                                               |                                        | 632.8                    | 0.132        |           |                                     | 72, W       |       |       |
| BTDA/DDS                                                                                |                                        | 632.8                    | 0.145        |           |                                     | 72, W       |       |       |
| Methylated humic acid                                                                   | THF                                    | 632.8                    | 0.22         | −0.042    |                                     | 73          |       |       |
| Benzylated humic acid                                                                   | THF                                    | 632.8                    | 0.19         | −0.041    |                                     | 73          |       |       |
| Hydroxyethyl cellulose acetate                                                          | THF                                    | 488                      | 0.0734       | 25        |                                     | 74          |       |       |
| Poly(vinyl acetate)                                                                     | THF                                    | 632.8                    | 0.0517       | 25        |                                     | 11          |       |       |
|                                                                                         |                                        |                          | 632.8        | 0.0471    | 26                                  |             | 20, X |       |
|                                                                                         |                                        |                          | 632.8        | 0.0402    | 26                                  |             | 20, Y |       |
|                                                                                         |                                        |                          | 632.8        | 0.077     | 23                                  |             | 20    |       |
|                                                                                         |                                        |                          | 546          | 0.104     | 25                                  |             | 10    |       |
|                                                                                         | ethyl acetate                          | 436                      | 0.104        | 25        |                                     | 10          |       |       |
|                                                                                         |                                        | acetone                  | 546          | 0.080     | 25                                  |             | 10    |       |
|                                                                                         |                                        |                          |              | 436       | 0.083                               | 25          |       | 10    |
|                                                                                         | MEK                                    |                          | 546          | 0.080     | 25                                  |             | 10    |       |
|                                                                                         |                                        |                          |              | 436       | 0.083                               | 25          |       | 10    |
|                                                                                         | Poly(vinyl alcohol)                    | 0.05 M NaNO <sub>3</sub> | 632.8        | 0.1501    | 27                                  |             | 75    |       |
|                                                                                         |                                        |                          |              | 632.8     | 0.1429                              | 27          |       | 75, Z |
|                                                                                         |                                        |                          | 632.8        | 0.1501    | 25                                  |             | 76, a |       |
|                                                                                         |                                        |                          | 632.8        | 0.1429    | 25                                  |             | 76, b |       |
| 0.1 M KH <sub>2</sub> PO <sub>4</sub>                                                   |                                        | 632.8                    | 0.1534       | 23        |                                     | 20          |       |       |
|                                                                                         |                                        | water                    | 546          | 0.164     | 30                                  |             | 10    |       |
|                                                                                         |                                        |                          |              | 436       | 0.168                               | 30          |       | 10    |
| Polyacrylamide                                                                          |                                        | water                    | 632.8        | 0.1735    |                                     |             | 77    |       |
|                                                                                         |                                        |                          |              | 632.8     | 0.142                               | 23          |       | 20, C |
|                                                                                         |                                        |                          |              | 632.8     | 0.1829                              | 25          |       | 20, d |
|                                                                                         |                                        |                          | 546          | 0.163     |                                     |             | 10    |       |
|                                                                                         |                                        |                          | 632.8        | 0.1587    | 25                                  |             | 20, e |       |
|                                                                                         | 1 M NaCl                               | 632.8                    | 0.1556       | 25        |                                     | 20, f       |       |       |
|                                                                                         | 1 M NaCl                               | 632.8                    | 0.163        |           |                                     | 78          |       |       |
|                                                                                         | 0.1 M Na <sub>2</sub> HPO <sub>4</sub> | 632.8                    | 0.176        | 23        |                                     | 79, g       |       |       |
|                                                                                         | 0.02 M Na <sub>2</sub> SO <sub>4</sub> |                          | 0.174        | 23        |                                     | 79, h       |       |       |
|                                                                                         |                                        |                          |              |           |                                     |             |       |       |
|                                                                                         | Poly(acrylamide-co-sodium acrylate)    | 1 M NaCl                 | 632.8        | 0.165     |                                     |             | 80    |       |
| 0.1 M NaH <sub>2</sub> PO <sub>4</sub> + 0.3 M Na <sub>2</sub> SO <sub>4</sub> , pH 2.2 |                                        |                          | 0.16         |           |                                     | 81          |       |       |
|                                                                                         |                                        |                          | 632.8        | 0.165     |                                     |             | 20    |       |
| (70/30)                                                                                 | 0.1 M NaNO <sub>3</sub>                | 632.8                    | 0.165        |           |                                     | 20          |       |       |
| Sodium polyacrylate                                                                     | 0.3 M NaCl                             | 488                      | 0.159        | 25        |                                     | 82          |       |       |
|                                                                                         | 0.1 M NaCl                             | 632.8                    | 0.175        | 23        |                                     | 20          |       |       |
|                                                                                         | water                                  | 632.8                    | 0.1463       | 23        |                                     | 20, i       |       |       |



| Sample                         | Solvent                                      | Wave-length (nm) | dn/dc (ml/g) | Temp (°C) | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|--------------------------------|----------------------------------------------|------------------|--------------|-----------|-------------------------------------|-------------|
| Poly(acrylic acid)             | aqueous NaCl                                 | 436              | 0.158        | 30        |                                     | 10          |
|                                |                                              | 632.8            | 0.0738       | 23        |                                     | 20          |
|                                |                                              | 632.8            | 0.0994       | 23        |                                     | 20          |
|                                | 0.2N HCl                                     | 632.8            | 0.133        |           |                                     | 83          |
| Poly(allyl amine)              | 0.1% TFA/                                    | 632.8            | 0.2034       | 35        |                                     | 84, j       |
|                                | 0.20M NaNO <sub>3</sub>                      | 632.8            | 0.1950       | 35        |                                     | 84, k       |
| Poly(vinyl amine)              | 0.1% TFA/0.20M NaNO <sub>3</sub>             | 632.8            | 0.1751       | 35        |                                     | 84, m       |
| Branched polyethyleneimine     |                                              |                  |              |           |                                     |             |
|                                | water                                        | 632.8            | 0.210        | 35        |                                     | 85          |
|                                | 1 M NaCl                                     | 632.8            | 0.195        | 35        |                                     | 85          |
|                                | methanol                                     | 632.8            | 0.225        | 35        |                                     | 85          |
| Poly(ethylene oxide)           | methanol                                     | 632.8            | 0.145        | 25        |                                     | 86          |
| Dextran                        | water                                        | 632.8            | 0.147        |           |                                     | 87, n       |
|                                | water                                        | 632.8            | 0.150        | 25        |                                     | 88          |
|                                | 0.1 M KNO <sub>3</sub>                       | 632.8            | 0.150        | 25        |                                     | 88          |
|                                | 0.05M KH <sub>2</sub> PO <sub>4</sub> (pH 7) | 632.8            | 0.137        |           |                                     | 87, n       |
|                                | 0.15M NaCl                                   | 632.8            | 0.145        |           |                                     | 89          |
|                                | 0.05M K <sub>2</sub> HPO <sub>4</sub> (pH 7) | 632.8            | 0.1378       |           | 0.41                                | 90          |
|                                | water                                        | 546              | 0.148        |           |                                     | 91          |
|                                | 0.5M NaCl                                    | 546              | 0.145        |           |                                     | 91          |
|                                | water                                        | 436              | 0.150        |           |                                     | 91          |
|                                | 0.5M NaCl                                    | 436              | 0.148        |           |                                     | 91          |
|                                | 0.2M NaNO <sub>3</sub>                       | 514.5            | 0.147        | RT        |                                     | 92          |
| DEAE-dextran                   | 0.8M NaNO <sub>3</sub>                       | 632.8            | 0.150        | 20        |                                     | 93, o       |
| Pullulan                       | 0.2M NaNO <sub>3</sub>                       | 514.5            | 0.147        | RT        |                                     | 92          |
|                                | 0.15M NaCl                                   | 632.8            | 0.145        |           |                                     | 89          |
|                                | water                                        | 488              | 0.148        | 25        |                                     | 94          |
| Polysaccharide                 | water                                        | 632.8            | 0.146        |           |                                     | 95          |
|                                | 0.1 M NaCl                                   | 632.8            | 0.143        |           |                                     | 96, p       |
| (anionic)                      | 0.1 M NaCl                                   | 632.8            | 0.143        |           |                                     | 96, q       |
| Sodium hyaluronate             | 0.1 M NH <sub>4</sub> NO <sub>3</sub>        | 632.8            | 0.155        |           | 3                                   | 97          |
| Sodium heparin                 | 0.1 M NaCl                                   | 632.8            | 0.163        | 23        |                                     | 98, r       |
| Sodium poly(styrene sulfonate) |                                              |                  |              |           |                                     |             |
|                                | water                                        | 632.8            | 0.17         |           |                                     | 99          |
| (0.9 meq/g)                    | 0.05 M NaCl                                  | 546              | 0.212        | 25        |                                     | 10          |
| (1.8 meq/g)                    |                                              | 546              | 0.207        | 25        |                                     | 10          |
| (2.0 meq/g)                    | 0.10 M NaCl                                  | 546              | 0.195        | 25        |                                     | 10          |
| (2.7 meq/g)                    |                                              | 546              | 0.205        | 25        |                                     | 10          |
|                                | 0.1 M NaNO                                   | 632.8            | 0.195        |           |                                     | 100         |
| Poly(sulfoalkyl methacrylate)  |                                              |                  |              |           |                                     |             |
| (ethyl)                        | 0.1 M NaCl                                   | 532              | 0.135        | 25        |                                     | 101         |
| (hexyl)                        |                                              | 532              | 0.140        | 25        |                                     | 101         |
| (decyl)                        |                                              | 532              | 0.170        | 25        |                                     | 101         |
| (octyl)                        |                                              | 532              | 0.149        | 25        |                                     | 101         |
| Hydroxyethyl-cellulose         | 0.1 M NaCl                                   | 632.8            | 0.159        |           |                                     | 102         |

| Sample                          | Solvent                                | Wave-length (nm) | dn/dc (ml/g) | Temp (°C)                | A <sub>2</sub> (× 10 <sup>3</sup> ) | Ref, Remark |
|---------------------------------|----------------------------------------|------------------|--------------|--------------------------|-------------------------------------|-------------|
| Carboxylmethyl-cellulose        | 0.1 M NaCl                             | 632.8            | 0.147        |                          |                                     | 102         |
| PNaAMPS                         | water                                  | 632.8            | 0.20         |                          |                                     | 99, s       |
| BSA                             | water                                  | 632.8            | 0.187        | 25                       |                                     | 103, t      |
| BSA-SDS                         | water                                  | 632.8            | 0.376        | 25                       |                                     | 103, u      |
| Ovalbumin-SDS                   | water                                  | 632.8            | 0.374        | 25                       |                                     | 103         |
| Ribonuclease-SDS                | water                                  | 632.8            | 0.398        | 25                       |                                     | 103         |
| Ribonuclease A                  | 75 % A – 25% B                         | 632.8            | 0.172        | 7.5 × 10 <sup>-2</sup>   |                                     | 104, v      |
| Lysozyme                        | 65 % A – 35% B                         | 632.8            | 0.182        | -5.39 × 10 <sup>-2</sup> |                                     | 104, v      |
| BSA                             | 58 % A – 42 % B                        | 632.8            | 0.172        | 7.49 × 10 <sup>-2</sup>  |                                     | 104, v      |
| Bovin alkaline phosphatase      | 50 % A – 50% B                         | 632.8            | 0.142        | 0.854                    |                                     | 105, w      |
|                                 | 60 % A – 40 % B                        | 632.8            | 0.1188       | -9.81                    |                                     | 105, w      |
|                                 | 70 % A – 30%B                          |                  | 0.0956       | -163                     |                                     | 106, w      |
|                                 | 70 % A – 30 % B                        | 632.8            | 0.0956       | -163                     |                                     | 105, w      |
| Bovin thyroglobulin (MW 669000) | 0.10 M Na <sub>2</sub> SO <sub>4</sub> | 632.8            | 0.165        | 20                       |                                     | 106, x      |
| Serum albumin (MW 68000)        | 0.10 M Na <sub>2</sub> SO <sub>4</sub> | 632.8            | 0.165        | 20                       |                                     | 106, x      |
| Chymotrypsinogen A (MW 25900)   | 0.10 Na <sub>2</sub> SO <sub>4</sub>   | 632.8            | 0.170        | 20                       |                                     | 106, x      |
| DNA                             | water                                  | 632.8            | 0.166        |                          |                                     | 99          |
| DNA fragment                    | 0.1 M NaCl                             | 632.8            | 0.168        | 25                       | 0.6                                 | 107         |

**Remarks.** A:  $dn/dc = 0.108 - 21.0/M_n$ ; B:  $dn/dc = 0.106 - 12.4/M_n$ ; C: polystyrene MW- $(dn/dc)_{\text{toluene}} - (dn/dc)_{\text{MEK}}$ ,  $1.8 \times 10^6 - 0.109 - 0.216$ ,  $97200 - 0.107 - 0.214$ ,  $51000 - 0.106 - 0.212$ ,  $19800 - 0.103 - 0.213$ ,  $10300 - 0.102 - 0.209$ ,  $4800 - 0.101 - 0.210$ ,  $2030 - 0.0985 - 0.208$ , polyethylene/trichlorobenzene  $dn/dc = -(0.1023 + (20.4/M))$ ; D:  $(dn/dc)_{670} = 0.9783 (dn/dc)_{632.8}$ ,  $(dn/dc)_{670} = 0.1804 - 9.149/M$ ; E: methylene chloride; F: DMF – *N,N*-dimethylformamide; G: 60% MMA diblock; H: PS/THF,  $A_2 = 0.01 M_w^{-0.25}$ , PMMA/THF,  $A_2 = 0.012 M_w^{-0.29}$ ; J: vinyl alcohol content ranging from 11 to 23%, hexafluoroisopropanol; K:  $M_w = 6160$ ; L:  $dn/dc = 0.111 - 15.3/M_n$  (36% *cis*-1,4, 57% *trans*-1,4, 7% 1,2-); M:  $M_w = 6100$  (0.0974 for  $M_w = 3300$ ); N: ethylene-co-octene-1; O: ethylene-co-butene-1; P: pentafluorophenol; Q: dichloroacetic acid; R:  $M_w - dn/dc$ , 35500 – 0.1417, 46200 – 0.1523, 63100 – 0.1477, 72400 – 0.1469; S: poly(methyl vinyl ether-co-maleic anhydride), hydroxymethyl aminomethane, pH 9; T: I – poly(diphenoxy)phosphazene, II – poly(aryloxy)phosphazene, III – poly(fluoroalkoxy)phosphazene, IV – poly[bis(trifluoroethoxy) phosphazene]; U: 6F-dianhydride-1,4-phenylenediamine  $dn/dc = 0.107$ , 6F-dianhydride-4,4-methylenediamine  $dn/dc = 0.132$ , benzophenone-3,3',4,4'-diaminodiphenylsulfone  $dn/dc = 0.145$ ; V: poly(arylether ether ketone); W: 6FDA – 6F-dianhydride, MDA – 4,4'-methylenediamine, DDS – 4,4'-diamine diphenyl sulfone, BTDA – benzophenone-3,3',4,4'-tetracarboxylic dianhydride, PDA – 1,4-phenylenediamine; X:  $M_w = 4.66 \times 10^6$ ; Y:  $M_w = 1.07 \times 10^5$ ; Z: 88% degree of hydrolysis; a: 98% hydrolysis; b: 88% hydrolysis; c:  $M_w = 5.5 \times 10^6$ ; d:  $M_w = 3.69 \times 10^6$ ; e:  $M_w = 7.72 \times 10^5$ ; f:  $M_w = 8 \times 10^6$ ; g:  $5.32 \times 10^5 M_w$ ; h:  $2.18 \times 10^6 M_w$ ; i:  $M_w = 28000$ ; j: TFA – trifluoroacetic acid,  $M_w = 3.61 \times 10^4$ ; k:  $M_w = 1.06 \times 10^5$ ; m:  $M_w = 2.50 \times 10^5$ ; n:  $M_w = 6.82 \times 10^4$ ; o: DEAE – 2-diethylaminoethyl; p: Schizophyllan

(neutral polysaccharide); q: xanthan (anionic heteropolysaccharide); r: 0.01 M NaCl – 0.142 –  $A_2 = 16$ , 0.1 M NaCl – 0.137 – 5.8, 0.5 M NaCl – 0.126 – 2.52; 1 M NaCl – 0.121 – 1.52; s: poly(sodium-2-(acrylamide)-2-methyl propanesulfonate); t: bovine serum albumin; u: sodium dodecyl sulfonate; v: A – 0.15% TFA in water, B – 0.15% TFA in 95% acetonitrile; w: A – 2.5 M ammonium sulfate in 0.5 M ammonium acetate (pH 6.0), B – 0.5 M ammonium acetate (pH 6.0); x: in 0.02 M  $\text{Na}_2\text{HPO}_4 - \text{NaH}_2\text{PO}_4$  at pH 6.9.

## References

1. ANDERSON RI (1976) *Am Lab* 8(5):95
2. HADJICHRISTIDIS N, FETTERS LJ (1982) *J Polym Sci Polym Phys Ed* 20:2163
3. JORDAN RC, SILVER SF, SEHON RD, RIVARD RJ (1984) In: PROVIDER T (ed) *Size Exclusion Chromatography*, ACS Symposium Series 245, American Chemical Society, Washington, DC, p 295
4. SEGUDOVIC N, KARASZ FE, MACKNIGHT WJ (1990) *J Liq Chromatogr* 13:2581
5. YUNAN W, ZHONGDE X, LI J, ROSENBLUM WM, MAYS JW (1993) *J Appl Polym Sci* 49:967
6. MALIHI FB, KUO CY, PROVIDER T (1984) *J Appl Polym Sci* 29:925
7. FRANK R, FRANK L, FORD NC (1995) In: PROVIDER T, BARTH HG, URBAN MW (eds) *Chromatographic Characterization of Polymers*, *Adv Chem Ser* 247, American Chemical Society, Washington, DC, p 110
8. WAGNER HL, HOEVE CAJ (1971) *J Polym Sci A-2*, 9:1763
9. KHAN HU, GUPTA VK, YAMIN M (1983) *J Macromol Sci – Chem* A20:503
10. BRANDRUP J, IMMERGUT EH (1972) *Polymer Handbook*, Wiley, New York
11. JORDAN RC, MCCONNELL ML (1980) In: PROVIDER T (ed) *Size Exclusion Chromatography*, ACS Symposium Series 138, American Chemical Society, Washington, DC, p 107
12. JENG L, BALKE ST, MOUREY TH, WHEELER L, ROMEO P (1993) *J Appl Polym Sci* 49:1359
13. MOUREY TH, MILLER SM, BALKE ST (1990) *J Liq Chromatogr* 13:435
14. LEE HC, REE M, CHANG T (1995) *Polymer* 36:2215
15. JENG L, BALKE ST (1993) *J Appl Polym Sci* 49:1375
16. MOUREY TH, COLL H (1995) In: PROVIDER T, BARTH HG, URBAN MW (eds) *Chromatographic Characterization of Polymers*, *Adv Chem Ser* 247, American Chemical Society, Washington, DC, p 123
17. OUANO AC, KAYE W (1974) *J Polym Sci Polym Chem Ed* 12:1151
18. MOUREY TH, BRYAN TG, GREENER J (1993) *J Chromatogr A* 657:377
19. KENT MS, TIRRELL M, LODGE TP (1994) *J Polym Sci Part B Polym Phys Ed* 32:1927 (1994)
20. Wyatt Technology, Data Sheet, Santa Barbara, CA
21. DUMELOW T (1989) *J Macromol Sci-Chem* A26:125
22. SUN SF, CHAO D, LIN S (1989) *Polym Commun* 24:84
23. FORGET JL, BOOTH C, CANHAM PH, DUGGLEBY H, KING TA, PRICE C (1979) *J Polym Sci Polym Phys Ed* 17:1403
24. AMBLER MR, MCINTYRE D (1977) *J Appl Polym Sci* 21:3237
25. LEE S, KWON O-S (1995) In: PROVIDER T, BARTH HG, URBAN MW (eds) *Chromatographic Characterization of Polymers*, *Adv Chem Ser* 247, American Chemical Society, Washington, DC, p 93
26. PODZIMEK S (1994) *J Appl Polym Sci* 54:91
27. BERKOWITZ SA (1983) *J Liq Chromatogr* 6:1359 (1983)
28. DEGOULET C, NICOLAI T, DURAND D, BUSNEL JP (1995) *Macromolecules* 28:6819
29. EVANS JM (1973) *Polym Eng Sci* 13:401
30. REMSEN EE (1991) *J Appl Polym Sci* 42:503
31. ASADUZZAMAN AKM, RAKSHIT AK, DEVI S (1993) *J Appl Polym Sci* 47:1813
32. MRKVICKOVA J, DANHELKA J (1990) *J Appl Polym Sci* 41:1929
33. HATTORI S, ENDOH H, NAKAHARA H, KAMATA T, HAMASHIMA M (1978) *Polym J* 10:173
34. RUDIN A, BENSCHOP-HENDRYCHOVA I (1974) *J Appl Polym Sci* 15:2881 (1971)

35. GEERISSEN H, ROOS J, WOLF BA (1985) *Makromol Chem* 186:735
36. CHEN X, XU Z, HADJICHRISTIDIS N, FETTERS LJ, CARELLA J, GRAESSLEY WW (1984) *J Polym Sci Polym Phys Ed* 22:777
37. HORSKA J, STEJSKAL J, KRATOCHVIL P (1983) *J Appl Polym Sci* 28:3873
38. HUBER CH, LEDERER K (1980) *J Polym Sci Polym Let Ed* 18:535
39. MRKVICKOVA L, LOPOUR P, POKORNY S, JANCA J (1980) *Angew Makromol Chem* 90:217
40. GRINSHPUN V, O'DRISCOLL KF, RUDIN A (1984) *J Appl Polym Sci* 29:1071
41. PANG S, RUDIN A (1993) In: PROVDER T (ed) *Chromatography of Polymers*, ACS Symposium Series 521, American Chemical Society, Washington, DC, p 254
42. WILD L, RANGANATH R, RYLE T (1971) *J Polym Sci A-2* 9:2137
43. BOEHN LL, LANVER U, LECHNER MD (1983) *Makromol Chem* 184:585
44. GRINSHPUN V, O'DRISCOLL KF, RUDIN A (1984) In: PROVDER T (ed) *Size Exclusion Chromatography*, ACS Symposium Series 245, American Chemical Society, Washington, DC, p 273
45. MACRURY TB, MCCONNELL ML (1979) *J Appl Polym Sci* 24:651
46. HORSKA J, STEJSKA J, KRATOCHVIL P (1979) *J Appl Polym Sci* 24:1845
47. RAM A, MILTZ J (1971) *J Appl Polym Sci* 35:2639
48. DROTT EE, MENDELSON RA (1970) *J Polym Sci Part A-2*, 8:1373
49. GRINSHPUN V, RUDIN A (1985) *Makromol Chem Rapid Commun* 6:219
50. BILLIANI J, LEDERER K (1990) *J Liq Chromatogr* 13:3013
51. GRINSHPUN, RUDIN A (1985) *J Appl Polym Sci* 30:2413
52. OGAWA T (1990) *J Liq Chromatogr* 13:51
53. BERKOWITZ S (1984) *J Appl Polym Sci* 29:4353
54. OHYOYA S, MATSUO T (1989) *Chem Expr (JPN)* 4:761
55. SCHORN H, KOSFELD R, HESS M (1983) *J Chromatogr* 282:579
56. MOUREY TH, BRYAN TG (1994) *J Chromatogr A* 679:201
57. TERAN CC, MACCHI EM, FIGINI RV (1987) *J Appl Polym Sci* 34:2433
58. OGAWA T, SAKAI M (1985) *J Polym Sci Polym Chem Ed* 23:1109
59. HAMMEL R, GERTH CH (1977) *Makromol Chem* 178:2697
60. SAMANTA SR (1992) *J Appl Polym Sci* 45:1635
61. JACKSON C, O'BRIEN JP (1995) *Macromolecules* 28:5975
62. PETROVIC ZS, MACKNIGHT WJ (1991) *Polym Bull* 27:281
63. PETRI HM, STAMMER A, WOLF BA (1995) *Makromol Chem Phys* 196:1453
64. WU CS, SENKA L, MALAWER EG (1989) *J Liq Chromatogr* 12:2919
65. SEDLACK J, VOHLIDE J (1993) *Makromol Chem Rapid Commun* 14:51
66. DE JAEGER R, LECACHEUX D, POTIN PH (1990) *J Appl Polym Sci* 39:1793
67. MOUREY TH, MILLER SM, FERRAR WT, MOLLAIRE TR (1989) *Macromolecules* 22:4286
68. KONAS M, MOY TM, ROGERS ME, SCHULTZ AR, WARD TC, McGRATH JE (1995) *J Polym Sci Part B Polym Phys* 33:1441
69. DEVAUX J, DELIMOY D, DAOUST D, LEGRAS R, MERCIER JP, NIELD E (1985) *Polymer* 26:1994
70. STACY CJ (1986) *J Appl Polym Sci* 32:3959
71. TALLEY CP, BOWMAN LM (1979) *Anal Chem* 51:2239
72. KONAS M, MOY TM, ROGERS ME, SHULTZ AR, WARD TC, McGRATH JE (1995) *J Polym Sci Part B Polym Phys* 33:1441
73. OLSON ES, DIEHL JW (1985) *J Chromatogr* 349:337
74. WU C, SIDDIG M, JIANG S, HUANG Y (1995) *J Appl Polym Sci* 58:1779
75. NAGY DJ (1995) In: WU C-S (ed) *Handbook of Size Exclusion Chromatography*, Dekker, New York, p 279
76. NAGY DJ (1986) *J Polym Sci Part C Polym Let Ed* 24:87
77. LESEC J, VOLET G (1989) *International GPC Symposium*, Massachusetts, p 386
78. LANGHORST MA, STANLEY FW Jr, CUTIE SS, SUGARMAN JH, WILSON LR, HOAGLAND DA, PRUD'HOMME RK (1986) *Anal Chem* 58:2242
79. KIM CJ, HAMIELEC AE, BENEDEK A (1982) *J Liq Chromatogr* 5:1277
80. MULLER G, YONNET C (1984) *Makromol Chem Rapid Commun* 5:197

81. PAPAZIAN LA (1990) *J Liq Chromatogr* 13:3389
82. KATO T, TOKUYA T, NOZAKI T, TAKAHASHI A (1984) *Polymer* 25:218
83. LACIK I, SELB J, CANDAN F (1995) *Polymer* 36:3197
84. NAGY DJ, TERWILLIGER DA, LAWREY BD, TIEDGE WF (1989) *International GPC Symposium, Massachusetts*, p 637
85. PARK IH, CHOI EJ (1996) *Polymer* 37:313
86. KINUGASA S, NAKAHARA H, KAWAHARA J, KOGA Y, TAKAYA H (1996) *J Polym Sci Part B Polym Phys* 34:583
87. LDC/Milton Roy/Chromatix Technical Note LS 7 (1979)
88. VAN DIJK JAPP, VARKEVISSER FA, SMIT JAM (1987) *J Polym Sci Part B Polym Phys* 25:149
89. JACKSON C, NILSSON L, WYATT PJ (1989) *J Appl Polym Sci Appl Polym Symp* 43:99
90. KIM CJ, HAMIELEC AE, BENEDEK A (1982) *J Liq Chromatogr* 5:425
91. VINK H, DAHLSTROM G (1967) *Makromol Chem* 109:249
92. BAHARY WS, HOGAN MP, JILANI M, ARONSON MP (1995) In: PROVIDER T, BARTH HG, URBAN MW (eds) *Chromatographic Characterization of Polymers*, *Adv Chem Ser* 247, American Chemical Society, Washington, DC, p 151
93. HANSELMANN R, BURCHARD W (1995) *Makromol Chem Phys* 196:2259
94. KATO T, OKAMOTO T, TOKUYA T, TAKAHASHI A (1982) *Biopolymers* 29:1623
95. YU LP, ROLLINGS JE (1988) *J Appl Polym Sci* 35:1085
96. CAPRON I, GRISEL M, MULLER G (1995) *Int J Polym Anal & Charact* 2:9
97. REED WF (1995) *Makromol Chem Phys* 196:1539
98. MIKLAUTS H, RIEMANN J, VIDIC HJ (1986) *J Liq Chromatogr* 9:2073
99. HOLZWARTH G, SONI L, SCHULZ DN (1986) *Macromolecules* 19:422
100. TIELKING H, KULICKE WM (1996) *Anal Chem* 68:1169
101. PEIGIANG W, SIDDIG M, HUIYING C, DI Q, WU C (1996) *Macromolecules* 29:277
102. PICTON L, MERLE L, MULLER G (1996) *Int J Polym Anal & Character* 2:103 (1996)
103. KONISHI K (1988) In: DUBIN P (ed) *Aqueous Size Exclusion Chromatography*, Elsevier, Amsterdam, p 327
104. MHATRE R, KRULL IS, STUTING HH (1990) *J Chromatogr* 502:21
105. KRULL IR, STUTING HH, KRZYSKO SC (1988) *J Chromatogr* 442:29
106. BINDELS JG, DE MAN BM, HOENDERS HJ (1982) *J Chromatogr* 252:255
107. NICOLAI T, VAN DIJK L, VAN DIJK JAPP, SMIT JAM (1987) *J Chromatogr* 389:286

---

## Subject Index

### A

absorbance 43  
additives 196  
adsorption 169  
–, coefficient 14  
–, effect 67  
agarose gel 47  
aggregate 160  
anionic polymers 170  
antioxidant 68  
aqueous SEC column 50–52, 65

### B

band broadening 90  
–, equation 90  
baseline  
–, selection 79  
–, fluctuation 69  
branched points 158  
– polyethylene 158  
branching factor 128  
– index 158  
– parameter 157  
Butylated hydroxytoluene (BHT) 155

### C

calculation of MW average 77  
calibration curve 62, 63, 95  
– standard 62  
capacity factor 25  
capillary viscometers 124  
cellulose nitrate 144  
chemical composition distribution 185  
– heterogeneity 187  
colligative properties 3  
columns 47  
– efficiency 24  
– selection 64, 65  
– selectivity 22, 25  
– temperature 73  
concentration dependence 58, 60  
– effect 58  
– -based detector 38  
conical model 14, 15  
copolymers 185

correction factor, MW 27  
– of peak broadening 90  
cumulative MWD 85, 86  
cylindrical pore model 15

### D

DALLS 118  
data handling system 89  
Debye plot 119  
deconvolution approaches 91  
deflection type/refractometer 39  
density detector 42  
dependence of retention volume on column  
temperature 73  
dextrans 46  
2,6-di-*tert*-butyl-4-methylphenol 155  
2,6-di-*tert*-butyl-*p*-cresol (BHT) 53  
differential MWD 85, 88  
– pressure 124  
–, transducer 124  
– refractometer 38  
distribution coefficient 16  
–, for size exclusion 14  
–, for solute-packing interaction effects 14  
Dondos–Benoit equation 137  
dual-angle laser LS 118

### E

effect of injection volume on retention  
69–71  
effective radius 103  
Einstein's viscosity equation 102  
end-group analysis 3  
end-to-end-distance 103  
–, of the chain 19  
enthalpic interactions 13  
epoxy resins 137–140  
equivalent hydrodynamic sphere 103  
error, MW 27  
ethylene–propylene copolymers 161, 186  
– -diene terpolymers 161  
ethylene-vinyl acetate copolymer 162  
evaporative light scattering detector 41  
exclusion limit 12, 64  
experimental conditions 69

- extended chain length 105–106  
extra-column band broadening 32
- F**  
flame ionization detector 42  
Flory constant 20  
– most probable distribution function 99  
– temperature 55  
– universal constant 20, 104  
Flory-Fox equation 20  
flow meters 83  
– rate dependence of retention volume 71  
flow-rate fluctuation 83  
Fourier-transform infrared (FTIR) spectro-  
photometer 44  
Fresnel's law 40
- G**  
Gaussian distribution 24  
–, function 85  
gel filtration chromatography (GFC) 4  
– permeation chromatography (GPC) 4  
geometric models 14  
Gladstone–Dale rule 121  
good solvent 55, 101
- H**  
*height equivalent to a theoretical plate*  
(HETP) 24  
hemicelluloses 181  
heparin 182  
high-temperature SEC 155  
–, (HTSEC) systems 33  
HPLC setup 32  
Huggins equation 108  
Huggins' constant 108  
hyaluronic acid 182  
hydrodynamic radius 103  
– volume 20, 61, 103, 126  
–, concept 17  
hydrogen bonding 169  
hydrophobic interaction 169
- I**  
infrared (IR) detector 44  
inherent viscosity 107  
injection volume 69–71  
injector 34  
instrumental spreading function 90  
integral MWD 85, 86–88  
– operator 91  
interferometer 40  
internal marker 83  
interstitial volume 11  
intramolecular electrostatic interaction 169  
intrinsic viscosity 20, 76, 108, 23  
ion exchange 168  
– exclusion 168  
–, effect 71  
– inclusion 168  
ionic strength 68  
Ionol 155  
Irganox 1010 156  
isoelectric point 172  
iterative methods 92
- K**  
Kraemer constant 108  
– equation 108
- L**  
lag time, detector 149  
LALLS 118  
light scattering 4  
linear calibration curve 97  
logarithmic-normal distribution 85  
long-chain branching 157–160  
low-angle laser LS 118
- M**  
MALLS 118  
Mark–Houwink constants 108, 148–149  
– equation 21, 108  
–, for copolymers 215  
–, for homopolymers 199  
–, for polar polymers 209  
measurement of flow rate 83  
melamine-formaldehyde resin 141  
membrane osmometry 3  
molar absorption coefficient 43  
molecular hydrodynamic volume 11  
– size 11, 101–105  
– weight distribution 85–88  
monomer unit 106  
multi-angle laser LS 118  
MW-sensitive detector 38
- N**  
narrow-molecular weight distribution  
standards 95  
non-aqueous SEC column 47–50  
– SEC columns 64–65  
nonionic polymers 169  
nonlinear calibration curve 98  
non-size exclusion effects 168  
non-solvent 55  
nucleic acids 180  
number of theoretical plates  $N$  24  
number-average molecular weight 1  
– MW 77

**O**

oligomers 131–137  
3rd-order polynomial 82  
overload effect 58

**P**

partition coefficient 14  
peak broadening 23, 26, 90, 93  
– variance 23  
peptides 175–177, 178–180  
pesticides 198  
phenol-formaldehyde resins 140  
photodiode array 43  
poly(arylether ether ketone) 164  
poly(2-vinylpyridine) 177  
poly(acrylonitrile) 147  
poly(ethylene terephthalate) 146  
poly(phenylene sulfide) 163  
poly(styrene-methyl methacrylate) 186  
poly(styrene(S)-acrylonitrile(AN))  
  copolymer 187  
poly(vinyl acetate) 147  
poly(vinyl alcohol) 147  
poly(vinyl butyral) 147  
poly(vinyl chloride) 142  
poly(vinyl pyridine) 147  
poly(2-hydroxy methacrylate gel 48  
polyacrylamide 174  
–, gel 48  
–, hydrolysate 177  
polyamic acid 148  
– ester 148  
polyamides 145–146  
polydisperse standards 97  
polydispersity 1, 78  
polyethylene 155–160  
–, glycols 131–134  
poly(glyceryl methacrylate) gel 47  
polyimide 148  
polymer solubility 55–57  
– standard 62  
polynomial 63, 91  
polyorganophosphazenes 144  
polypropylene 155, 160–161  
polysaccharides 180  
polystyrene (PS) gel 47  
– sulfonate gel 48  
poly(vinyl alcohol) gel 48  
polyvinylpyrrolidone 174  
poor solvent 55, 101  
pore size of PS gel 64  
– volume 12  
preferred sample concentration 62  
preparation of mobile phase 69  
preparative SEC 189–194

primary standard 102

proteins 172–173, 175–177, 178–180

PS standard 63

pumps 34–37

**Q**

Q-factor 106–107

**R**

radius of gyration 18, 103, 104  
RALLS 118, 128–129  
Rayleigh ratio 115  
reciprocal slope 22  
recycle SEC 191–194  
reduced viscosity 20, 107  
refractive index 38, 131–134  
–, increments 121–122  
–, (RI) detector 38–41  
relative viscosity 20, 107, 123  
resolution 22, 23, 26  
response correction factor 132  
retention volume 12, 21  
right-angle laser LS 118, 128–129  
root-mean-square distance 18  
round-robin experiments 82

**S**

sample concentration 70–71  
scattering angle 115  
Schulz-Zimm distribution 85  
SEC distribution coefficient 12  
secondary standard 100–101  
selection of columns 64–66  
– of mobile phase 68–69  
selectivity 22  
–, factor 25  
semirigid polymer gels 47  
separation mechanism 14  
– performance 64  
sequential U-test 150–152  
short-chain branching 157  
silica-based packing materials 46  
size exclusion chromatography (SEC) 4  
slope of the calibration curve 22  
sodium azide 69  
Sodium poly(acrylate) 177  
– poly(styrene sulfonate) 170, 174  
soft polymer gels 47  
Solomon-Ciuta equation 108  
solubility of a polymer sample 55–57  
– parameter 56  
specific refractive index increment  
  121–122, 217  
– resolution 26  
– viscosity 20, 107, 125



starch hydrolysate 180  
Stokes–Einstein equation 104  
Stokes' radius 103  
structure-selective detectors 43–44  
system peaks 13

**T**  
TALLS 118  
thermodynamic model 16  
theta solvent 55, 59, 60, 108  
– temperature 55  
third-order polynomial 96  
total column volume 22  
– mobile phase volume 12  
– permeation limit 12  
– permeation volume 12  
– pore volume 12  
triple-angle laser LS 118  
Tung's equation 90  
ultraviolet (UV) and visible (VIS)  
absorption detectors 43  
universal calibration curve 109–113, 126  
unperturbed root-mean-square end-to-end  
distance 20

**V**  
vapor pressure osmometry 3  
variances 23  
viscosity average MW 78  
– effect 60  
viscous fingering 58  
void volume 11

**W**  
weight-average 77  
–, molecular weight 1

**X**  
xanthan 182

**Z**  
( $z + 1$ )-average MW 78  
z-average mean-square radius of gyration  
103  
– molecular weight 1, 77  
Zimm plot 116